Notebook on yeast gene knockout experiments

No 1 Date: April 29 Experimenters: Chengxi Luo, Yuquan Luo Recorder: Chengxi Luo

Experiment	*E. coli* Top10 (pRS415) revival and culture
Note	Steps refer to Protocol 1 (*E. coli* Transformation and Culture Expansion) in the protocol. Key params: inoculate 10 μL of the -20 °C glycerol stock into 5 mL LB + Amp (100 ug/mL); culture at 37 °C, 220 rpm; inoculated at 22:31 on 2026-04-29. Parameter difference: Protocol 1 illustrates pRS426 transformation at 20 ng/μL; here the pRS415-carrying glycerol stock is directly recovered without transformation.
Result	The overnight culture turned visibly turbid after ~19 h of shaking, indicating sufficient biomass.
Explanation	The glycerol stock was recovered successfully with normal growth, meeting the input needed for downstream plasmid extraction.
Next plan	Harvest the culture the next day (2026-04-30) and proceed to plasmid extraction (Protocol 3).

No 2 Date: April 30 Experimenters: Chengxi Luo, Yuquan Luo Recorder: Chengxi Luo

Experiment	pRS415 plasmid extraction and QC
Note	Steps refer to Protocol 3 (Plasmid Extraction from *E. coli*) in the protocol. Key params: 4 tubes x 1.5 mL culture each; P1/P2/P3 = 250/250/350 µL; 500 µL BL balance; 600 µL PW wash x2; elute with 50 µL sterile water pre-heated to 65 °C; measure by NanoDrop. Harvested at 17:55 (total ~19 h 24 min). Parameter difference: elution volume is 50 µL here (to concentrate the plasmid); if the kit default is 100 µL, the actual 50 µL value prevails.
Result	Four samples obtained (prs1-prs4, ~50 µL each). A260/A280 = 1.864-1.921 (ideal 1.8-2.0), A260/A230 = 2.283-2.424 (>2.0), concentration 112.50-138.45 ng/µL. Purity good.

Table 1 Nanodrop readings of prs1-prs4

Sample	DNA (ng/uL)	A260/A280	A260/A230
prs1	138.45	1.900	2.309
prs2	112.50	1.918	2.368
prs3	114.95	1.921	2.283
prs4	115.00	1.864	2.424

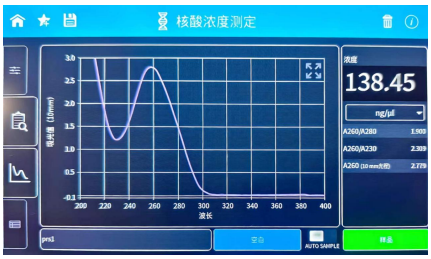


Fig. 1 prs1 Nanodrop

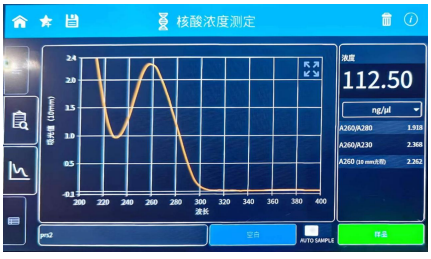


Fig. 2 prs2 Nanodrop

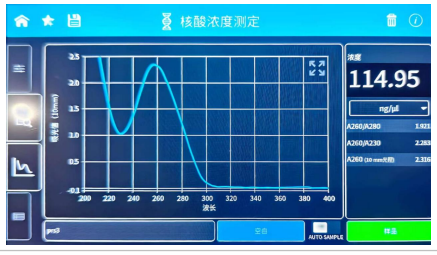
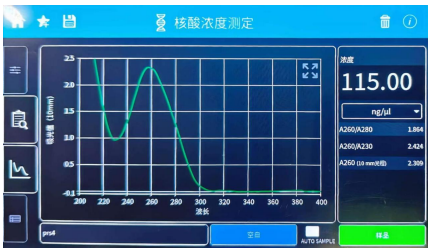


	Fig. 3 prs3 Nanodrop  **Fig. 4 prs4 Nanodrop**
Explanation	No obvious protein, salt or ethanol contamination; plasmid purity is good with sufficient total yield (~24 ug) for subsequent PCR.
Next plan	Use the purified plasmids as template to design and amplify the reporter gene; supplement agarose-gel electrophoresis results; store plasmids at -20 °C avoiding repeated freeze-thaw.

No 3 Date: May 1 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	BY4741 overnight culture
Note	Key params: inoculate one single colony into 5 mL YPD; 30 °C shaker, overnight; used on 2026-05-02 for extraction.
Result	The overnight culture turned visibly turbid; biomass met the extraction demand (no quantitation this day).
Explanation	No contamination or obvious sediment observed; culture was used on 2026-05-02 for snailase concentration optimization.
Next plan	On 2026-05-02, carry out snailase-concentration optimization (50/100/200/400 mg/mL) for yeast genomic DNA extraction.

No 4 Date: May 2 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	Snailase dosage optimization
Note	Steps refer to the magnetic-bead yeast genomic DNA extraction protocol (Protocol 23). Fixed: incubation 90 min (37 °C metal bath), 200 µL Reaction Buffer (sorbitol buffer); snailase 50/100/200/400 mg/mL, each with two single colonies (rep 1/rep 2). Main params: collect 1 mL culture; wash 500 µL PBS; 200 µL Reaction Buffer + 20 µL snailase + 5 µL 10 mg/mL RNase A (37 °C); proteinase K 20 µL + lysis buffer 500 µL (65 °C); 450 µL isopropanol + 20 µL beads; wash 600 µL Wash I then Wash II; elute 100 µL at 65 °C; measure by nucleic-acid/protein analyzer.
Result	DNA yield rose as snailase concentration increased; value after "/" is rep 2. A260/A280 ≈ 2.0 for all groups; A260/A230 mostly below 2.0.

Table 1 Snailase-concentration optimization (rep1 / rep2)

Snailase (mg/mL)	DNA (ng/uL)	A260/A280	A260/A230
50	920 / 1045	2.010 / 2.047	1.053 / 1.196
100	1246.9 / 1021.8	2.119 / 2.135	1.622 / 1.111
200	1284.7 / 1048.1	2.182 / 2.153	1.498 / 1.189
400	1323.9 / 1174.6	2.173 / 2.197	1.440 / 1.392

Explanation	All groups had A260/A280 near 2.0 with good overall purity. Yield increased with snailase concentration, highest at 400 mg/mL but already saturating at 100 mg/mL; some A260/A230 < 2 suggest residual salt/sugar - to be addressed via incubation time and buffer (see 05-03).
Next plan	On 2026-05-03, optimize 37 °C incubation time (30/60/90/120 min) and buffer type (Reaction Buffer/sorbitol vs PBS) with 50/100 mg/mL snailase.

No 5 Date: May 3 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	Incubation time and buffer optimization
Note	Steps same as 2026-05-02 (magnetic-bead yeast genomic DNA extraction, Protocol 23). Fixed: 175 µL buffer; incubation 30/60/90/120 min; snailase 50/100 mg/mL. In the tables, value before/after "/" = 50(before)/100(after) mg/mL.
Result	Detection results are summarized in Table 1 and Table 2 (value before/after "/" = 50 mg/mL / 100 mg/mL snailase).

time (min)	DNA (ng/uL)	A260/A280	A260/A230
30	1214.6 / 998.95	2.194 / 2.168	3.250 / 2.689
60	1186.5 / 1212.0	2.251 / 2.159	5.060 / 4.858
90	1079.0 / 1057.5	2.248 / 2.252	1.302 / 1.136
120	1243.5 / 1266.5	2.273 / 2.274	1.324 / 1.363

Table 1 175 µL Reaction Buffer (sorbitol)

time (min)	DNA (ng/uL)	A260/A280	A260/A230
30	801.00 / 915.8	2.156 / 2.173	10.14 / 2.278
60	1044.5 / 1024.5	2.283 / 2.313	7.130 / 3.888
90	957.0 / 681.95	2.233 / 2.324	1.246 / 1.056
120	944.0 / 946.70	2.145 / 2.157	1.118 / 0.831

Table 2 175 µL PBS

Explanation	Reaction Buffer gave higher DNA yield than PBS; purity (A260/A280 2.1-2.3) was good for both. A260/A230 was abnormally high at short times (e.g. PBS-50 mg/mL 10.14, Reaction Buffer-60 min 5.06) and normalized after 60 min (mainly 1-3), likely due to incomplete elution or baseline - to recheck. Best combination: Reaction Buffer (sorbitol) + 100 mg/mL snailase + 60 min incubation.
Next plan	Proceed with formal genomic DNA extraction under the optimal condition (Reaction Buffer + 100 mg/mL snailase + 60 min) and use the product for downstream experiments.

No 6 Date: May 8 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	*E. coli* Top10 (pESC-HIS-yEGFP) revival and culture
Note	Steps refer to Protocol 1 (*E. coli* Transformation and Culture Expansion) in the protocol. Key params: inoculate 10 μL of the -20 °C glycerol stock into 5 mL LB + Amp (100 ug/mL); culture at 37 °C, 220 rpm; inoculated at 21:35 on 2026-05-08. Parameter difference: the pESC-HIS-yEGFP-carrying glycerol stock is directly recovered without a transformation step.
Result	The overnight culture turned visibly turbid after ~18 h of shaking; biomass was sufficient (no quantitation this day).
Explanation	The glycerol stock was recovered successfully with normal growth, meeting the input needed for plasmid extraction.
Next plan	Harvest the culture the next day (2026-05-09) and proceed to plasmid extraction (Protocol 3).

No 7 Date: May 9 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	pESC-HIS-yEGFP plasmid extraction and QC
Note	Steps refer to Protocol 3 (Plasmid Extraction from *E. coli*) in the protocol. Key params: 4 tubes x 1.5 mL culture each; P1/P2/P3 = 250/250/350 µL; 500 µL BL balance; 600 µL PW wash x2; elute with 50 µL sterile water pre-heated to 65 °C; measure by NanoDrop. Harvested at 15:38 (total ~18 h 3 min). Parameter difference: elution volume is 50 µL here (to concentrate the plasmid); if the kit default is 100 µL, the actual 50 µL value prevails.
Result	Four samples obtained (yEGFP1-yEGFP4, ~50 µL each). A260/A280 = 1.859-1.921 (ideal 1.8-2.0), A260/A230 = 1.544-1.741 (below 2.0), concentration 34.950-44.850 ng/µL (avg 41.1), total ~8.2 ug.

Table 1 Plasmid concentration & purity of yEGFP1-yEGFP4

Sample	DNA (ng/uL)	A260/A280	A260/A230	A260 (10 mm)
yEGFP1	44.850	1.921	1.544	0.906
yEGFP2	34.950	1.859	1.660	0.715
yEGFP3	44.750	1.888	1.664	0.905
yEGFP4	39.950	1.862	1.741	0.816

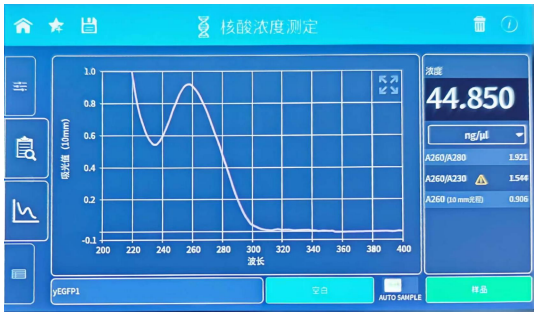


Fig. 1 yEGFP1 Nanodrop

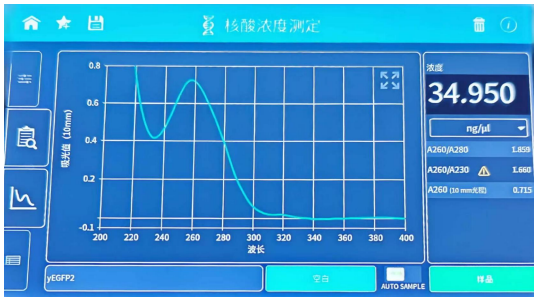
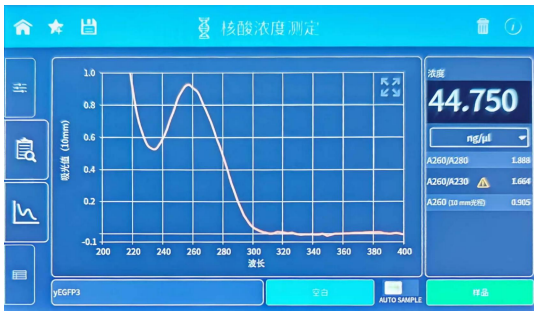
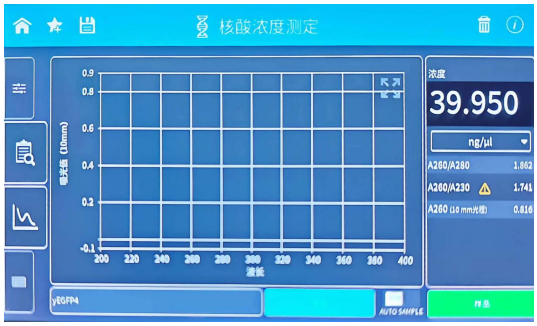


Fig. 2 yEGFP2 Nanodrop

	 **Fig. 3 yEGFP3 Nanodrop**  **Fig. 4 yEGFP4 Nanodrop**
Explanation	A260/A280 within 1.8-2.0 indicates little protein contamination and good purity; A260/A230 below 2.0 suggests residual salt/ethanol - dilute or re-precipitate before use. Total yield (~8.2 ug) is sufficient for downstream double digestion.
Next plan	Use the purified plasmids for BamH I + EcoR I double digestion to verify *yEGFP* insertion and PCR linearization; supplement agarose-gel results; store plasmids at -20 °C avoiding repeated freeze-thaw.

No 8 Date: May 9 Experimenters: Chengxi Luo Recorder: Chengxi Luo

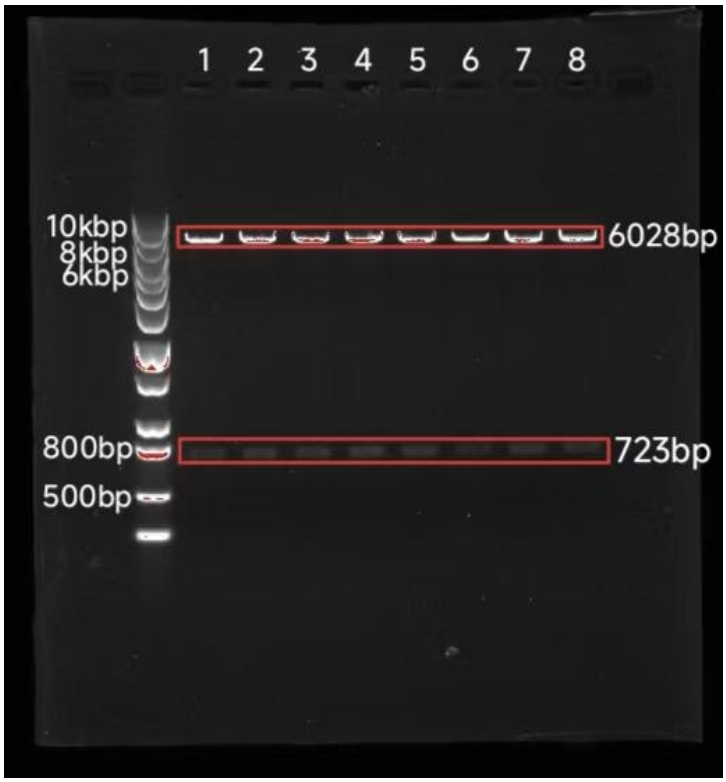
Experiment	pESC-HIS-yEGFP double-digestion check
Note	Steps refer to Protocol 4 (Restriction Enzyme Digestion) and Protocol 5 (Agarose Gel Electrophoresis) in the protocol. Key params: per tube, 10 µL double-digest master mix + 5 µL plasmid; master-mix order on ice (nuclease-free water -> 10x RE buffer -> BamH I + EcoR I), aliquoted to tubes 1-8, 37 C for 1 h then on ice; 1% agarose (0.5 g/50 mL 1x TAE) + 5 µL GelRed, 5 µL product + 1 µL 6x loading buffer plus DNA Marker, then imaged. Parameter difference: BamH I + EcoR I double digestion at 37 C for 1 h; gel run at 80 V for 1 h (Protocol 5 default about 90 V, 50 min).
Result	All double-digestion samples show bands consistent with the expected size; the recombinant-plasmid digestion verification is fully normal. Plasmid concentration detection is shown below.  **Fig. 1 pESC-HIS-yEGFP BamH I / EcoR I double-digestion gel**

	 **Fig. 2 Nucleic-acid concentration of pESC-HIS-yEGFP plasmid**
Explanation	The recombinant plasmid is validated; this batch of plasmids can serve as the template for PCR amplification of the linearized vector.
Next plan	Use the validated plasmids as template for PCR amplification of the linearized vector, and prepare template DNA for the *lacZα* / *FUS1* amplifications.

No 9 Date: May 10 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	*lacZα* and *FUS1* promoter PCR
Note	Steps refer to Protocol 7 (Conventional PCR) and Protocol 5 (Agarose Gel Electrophoresis) in the protocol. Key params: *lacZα* (50 µL): template 2 µL, 0.4 µM each primer, 25 µL 2x mix; 94 C 3 min, 35 cycles (94 C 30 s / 52 C 30 s / 72 C 20 s), final 72 C 5 min. *FUS1* (50 µL): template 1 µL, other components as above; 94 C 3 min, 35 cycles (94 C 30 s / 55 C 30 s / 72 C 15 s), final 72 C 10 min. Products run on 1% agarose, imaged. Parameter difference: standard-Taq 94 C program, 35 cycles; annealing set primer-specifically (*lacZα* 52 C, *FUS1* 55 C) and extension shortened for the short fragments (*lacZα* 20 s, *FUS1* 15 s), shorter than the ~1 kb/min rate implied by Protocol 7; gel run 80 V, 1 h.
Result	*lacZα* (Fig. 1): the target-size band is visible but with some non-specific bands. *FUS1* (Fig. 2): only samples 1 and 3 amplified the expected band, and all 4 samples show extra bands. *FUS1* template DNA concentrations are listed in Table 1. *lacZα*（

Sample	Template DNA (ng/uL)
FUS1-1	1044.5
FUS1-2	1024.5
FUS1-3	1044.5
FUS1-4	1024.5

Table 1 Template DNA concentration of *FUS1* samples

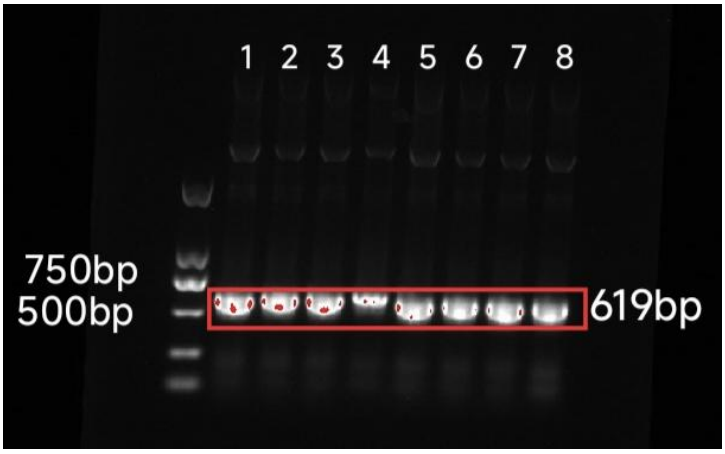


Fig. 1 *lacZα* homology-arm fragment PCR gel

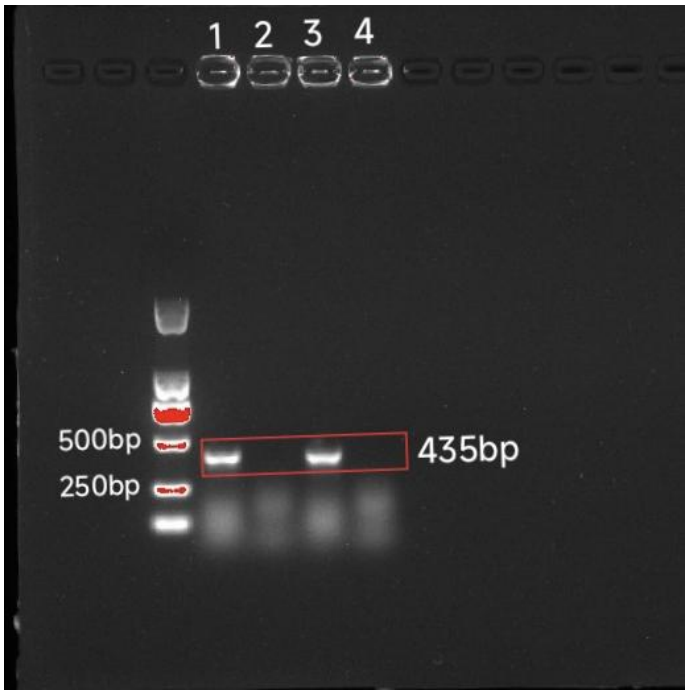


Fig. 2 *FUS1* promoter first-round PCR gel

样品	浓度 (ng/μL)
prs1	138.45
prs2	112.50
prs3	114.95
prs4	115.00

Fig. 3 Nucleic-acid concentration of pRS415 plasmid

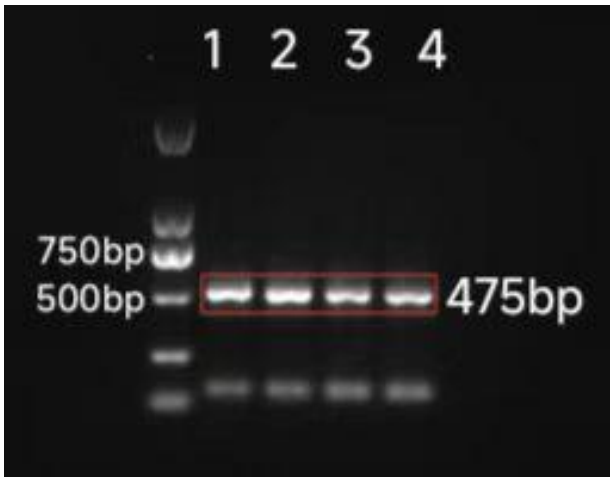
Explanation	*lacZα* shows non-specific bands; *FUS1* was amplified successfully only in samples 1 and 3, and the template-concentration difference may have affected the outcome.
Next plan	Raise the annealing temperature, reduce the cycle number, or gel-purify the target band for *lacZα*. Use the *FUS1* samples 1 and 3 products as template for a second PCR with homology arms.

No 10 Date: May 11 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	Linearized-vector and *FUS1* second-round PCR
Note	Steps refer to Protocol 7 (Conventional PCR, high-fidelity program) and Protocol 5 (Agarose Gel Electrophoresis) in the protocol. Key params: Linearized vector (50 µL, high-fidelity): template 5 µL, 0.4 µM each primer, 25 µL 2x mix; 98 C 5 min, 30 cycles (98 C 10 s / 55 C 15 s / 72 C 4 min), final 72 C 7 min; template numbers match No.8 Fig.2 samples. *FUS1* second-round (50 µL, high-fidelity): 1:100-diluted product 1 µL as template; 98 C 3 min, 30 cycles (98 C 10 s / 55 C 15 s / 72 C 30 s), final 72 C 5 min; tubes 1-2 use product #1, tubes 3-4 use product #3. Products run on 1% agarose, imaged. Parameter difference: inverse-PCR linearization of the ~6.7 kb vector uses 98 C 5 min initial denaturation and 72 C 4 min extension; second-round *FUS1* template diluted 1:100 to reduce impurity carry-over; annealing 55 C and gel run 80 V, 1 h (adjusted per Protocol 7/5).
Result	Linearized-vector PCR (Fig. 1): a target band is observed in all samples but accompanied by non-specific bands; sample numbers correspond one-to-one with the plasmids in No.8 Fig. 2. *FUS1* homology-arm second-round PCR (Fig. 2): samples 1-4 completed; templates of samples 1/2 and 3/4 come from first-round products #1 and #3, respectively.



Fig. 1 Linearized pESC-HIS-yEGFP vector PCR gel

	 **Fig. 2 *FUS1* promoter homology-arm second-round PCR gel**
Explanation	The linearized-vector PCR shows clear non-specific amplification; diluting the original PCR template reduced interference from impurities in the crude product.
Next plan	Purify the linearized vector, *lacZα* and homology-arm *FUS1* fragments to reduce the interference of extra fragments, then use the purified fragments for yeast homologous-recombination assembly.

No 11 Date: May 12 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	PCR purification and Gibson assembly
Note	Steps refer to Protocol 9 (PCR Purification), Protocol 11 (Gibson Assembly) and Protocol 5 (Agarose Gel Electrophoresis) in the protocol. Key params: Spin-column cleanup (Wash Solution + 80% ethanol, Buffer B3 + 20% isopropanol): 40 µL product made up to 100 µL + 5x B3, bind 8000 x g 30 s; wash x2 with 500 µL Wash Solution at 9000 x g 30 s; dry 9000 x g 1 min; elute with 40 µL Elution Buffer pre-warmed to 60 C, stand 2 min, centrifuge 1 min (conc./purity in Table 1). Gibson 10 µL mix (on ice): 5.0 µL 2x Master Mix (T5 exonuclease + Phusion polymerase + Taq ligase) + 1.0 µL linearized vector (~72.95 ng, 0.0164 pmol) + 0.76 µL *FUS1* insert (~10.3 ng, 0.0328 pmol) + nuclease-free water to 10 µL, then 37 C 30 min (Table 2); loading-mass conversion m(ng)=n(pmol) x L(bp) x 0.66. Identification gel: 1% agarose, 1x TAE, 5 µL sample + 1 µL 6x loading buffer, 80 V 60 min. Parameter difference: Gibson assembly run at 37 C for 30 min (Protocol 11 default 50 C, 15 min); elution volume 40 µL (within Protocol 9 range 30-50 µL); identification gel 80 V 60 min (Protocol 5 default about 90 V, 50 min).
Result	Spin-column recoveries (Table 1): linearized vector A, 72.950 ng/µL, A260/A280=1.838, A260/A230=1.974 (single ~6.7 kb band); *FUS1* insert B, 13.500 ng/µL, A260/A280=1.698, A260/A230=0.612 (single ~475 bp band, purity low). Gibson mix prepared correctly; the 37 °C 30 min reaction completed. Assembly gel (Fig. 1): Marker shows 15/10/7/5/2.5 kb ladder; linearized vector (lanes 6-7) gives a single bright ~6.7 kb band between 7-10 kb; the original circular plasmid (lanes 4-5) migrates differently due to supercoiled/open-circle conformations; Gibson products (lanes 2-3) enrich at the well/high-MW region consistent with a circular plasmid and are clearly distinct from the linearized single band, indicating re-circularization to a ~7.2 kb recombinant plasmid with the un-ligated linear band largely gone. Top-well smearing may reflect large ligation products/aggregates and higher loading; colony PCR and sequencing are required for final confirmation.

Sample	Fragment	Elution (uL)	DNA (ng/uL)	A260/A280	A260/A230	Gel band
A	Linearized vector	50	72.950	1.838	1.974	~6.7 kb single
B	FUS1 insert	50	13.500	1.698	0.612	~475 bp single

Table 1 Spin-column recovery products

Component	L (bp)	n (pmol)	m (ng)	C (ng/uL)	V (uL)
Linearized vector pESC-HIS-yEGFP	6746	0.0164	72.95	72.950	1.0
FUS1 insert (homology arms)	475	0.0328	10.3	13.500	0.76

Table 2 Gibson loading molar-ratio conversion (vector:insert 1:2)

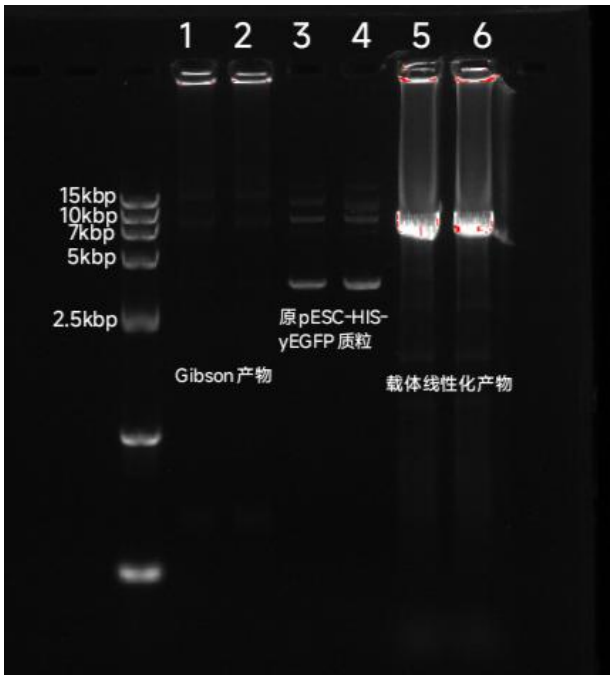


Fig. 1 Gibson products vs original plasmid & linearized vector, 1% agarose gel (lane 1 Marker; 2-3 Gibson; 4-5 original; 6-7 linearized)

Explanation	Purification removes primers, primer dimers and soluble impurities but has limited ability to clear fragments close in size to the target; the linearized large vector (6.7 kb) gives low self-ligation background, and 'vector-only'/'insert-only' negative controls can further assess self-ligation and recombination efficiency. Insert purity is low (A260/A230=0.612, salt/ethanol carry-over), but the ng-level loading mass is sufficient for the assembly.
Next plan	Transform BY4741 with the Gibson product on deficiency-selectable plates, pick single clones for colony PCR, send positives for sequencing to confirm the insertion orientation and sequence of the *FUS1* promoter in the pESC-HIS-yEGFP MCS; prepare glycerol stocks of verified BY4741 for functional assays.

No 12 Date: May 25-May 26 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	Competent-cell prep and transformation
Note	Steps refer to Protocol 12 (Yeast Transformation, incl. competent-cell preparation) and Protocol 14 (Yeast Growth Curve Measurement) in the protocol. Key params: BY4741 is his3-, the pESC-HIS backbone carries HIS3, so only plasmid-containing cells grow on SD-HIS (background control = wild-type BY4741 plated directly); 50 mL culture grown to OD600 0.4-0.6; competent cells split into 3 x 100 µL tubes transformed with 2, 5 and 7 µL Gibson product, all with Buffer C / DMSO heat shock; SD-HIS plates incubated inverted at 30 C from 05-26 23:34 to 06-01 19:49. Parameter difference: transformed with unpurified Gibson product (2/5/7 µL) rather than a defined plasmid amount; carrier DNA freshly denatured (100 C 15 min, ice >=15 min) to keep single-stranded; competent cells handled gently with pre-warmed Buffer A/B/C.
Result	Growth & OD600 (Table 1): the 50 mL culture rose from 0.2513 (15:43) to 0.4623 (20:49, +4 h 21 min), reaching the 0.4-0.6 window for competent-cell prep. Competent cells & grouping: 3 tubes of 100 µL, transformed with 2, 5 and 7 µL Gibson product respectively, all completing the Buffer C / DMSO heat-shock.

Time	Culture & operation	Temp · Speed	OD600	Note
05-26 15:43	Transfer to 50 mL YPD	30 C · 220 rpm	0.2513	Right after transfer; in 0.2-0.3 start range
05-26 20:49	50 mL YPD (+4 h 21 min)	30 C · 220 rpm	0.4623	Reached 0.4-0.6; for competent-cell prep

Table 1 BY4741 growth-curve / OD600 record

Plating (Fig. 1): SD-HIS plates incubated at 30 °C inverted from 05-26 23:34 to 06-01 19:49 (~5 d 20 h). Plate A (negative control) showed no colonies; plates B and C carried only a few pinpoint-to-punctate cream-white colonies - sparse and extremely tiny. Transformation occurred but efficiency was clearly low, and colony number showed no clear gradient with input volume.

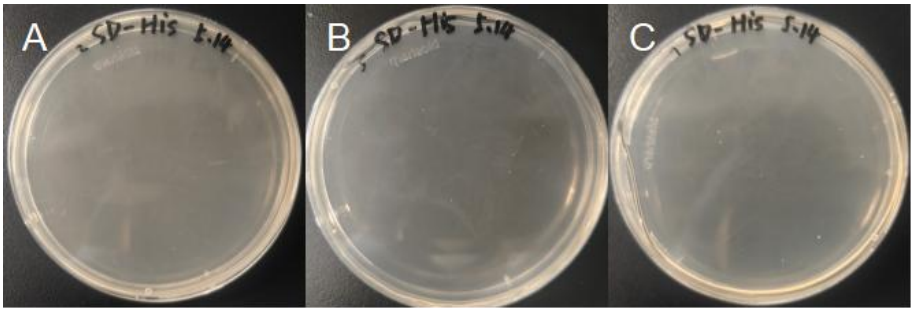


Fig. 1 SD-HIS screening plates after transforming BY4741 with Gibson products (plated 05-26 23:34, 30 °C inverted to 06-01 19:49, ~5 d 20 h): A = negative control, no single colonies; B and C = transformations, only a few extremely tiny colonies

Explanation	Low colony count likely reflects: residual salts / enzymes in the Gibson reaction inhibiting transformation; the ligation product being predominantly linear or nicked rather than circular; insufficient competent-cell efficiency; and the low purity of the *FUS1* fragment (05-12 measured A260/A230 only 0.612).
Next plan	Re-purify the Gibson product by spin column (or concentrate the competent cells / increase input) and re-transform; pick the colonies already present to YPD to amplify; confirm *FUS1* orientation and integrity by colony PCR / sequencing; and include wild-type BY4741 plated directly on SD-HIS as a background control.

No 13 Date: June 1-June 3 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	Positive transformant liquid culture
Note	On 1 June, four single colonies were picked from the positive SD-HIS transformation plates (No. 12, Fig. 1) and inoculated into 5 mL SD-HIS liquid medium. The culture was set up at 21:55 on 1 June and harvested at 16:18 on 3 June. Because the colonies were very small and the selection plates had already been incubated for about 6 days, the cells were of low viability and the expansion took longer than usual. The expanded cultures were reserved for the subsequent verification PCR. The expansion was carried out at 30 °C with shaking at 250 rpm.
Result	The expanded cultures turned visibly turbid, indicating sufficient biomass.
Explanation	The prolonged growth is consistent with the low viability of the transformants: the picked colonies were tiny, and the SD-HIS plates had been incubated for about 6 days before picking, so part of the inoculated cells were stressed and their lag phase was extended.
Next plan	The expanded cultures will be used as templates for the verification PCR to confirm integration of the construct.

No 14 Date: June 4 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	Transformant verification PCR (NaOH lysis)
Note	Steps refer to Protocol 13 (Colony PCR for *Saccharomyces cerevisiae*, NaOH-lysis method) and Protocol 5 (Agarose Gel Electrophoresis) in the protocol. Key params: 1 mL of expanded culture (No.13) centrifuged 12000 rpm 1 min, supernatant discarded; pellet lysed in 100 µL 0.2 M NaOH at 98 C for 10 min; 4.6 µL crude lysate added to a 10 µL verification PCR mix (94 C 3 min; 94 C 30 s / 55 C 30 s / 72 C 30 s x 30; 72 C 5 min; hold 4 C); products run on agarose gel, imaged. Parameter difference: lysis used 0.2 M NaOH at 98 C for 10 min, far stronger than Protocol 13 mild 20 mM NaOH lysis - this over-alkaline condition is the likely cause of template degradation and PCR failure; verification-PCR extension 30 s for a short locus; gel run 80 V.
Result	No target band of the expected size was detected on the gel (Fig. 1); only faint smearing / non-specific products were visible.  **Fig. 1 Agarose gel electrophoresis of the verification PCR products from the NaOH lysates: the leftmost lane, DNA marker; the remaining lanes showed no band of the expected size.**
Explanation	The absence of any specific product is most likely due to the too-strong alkalinity of the lysis reagent: 0.2 M NaOH at 98 °C may have degraded the template DNA and/or inactivated the DNA polymerase carried over into the reaction, so amplification failed completely.
Next plan	The alkaline lysis protocol will be optimised (lower NaOH concentration, shorter lysis time or thorough neutralisation); a milder snailase lysis (Protocol 23) was applied on 5 June (No. 15).

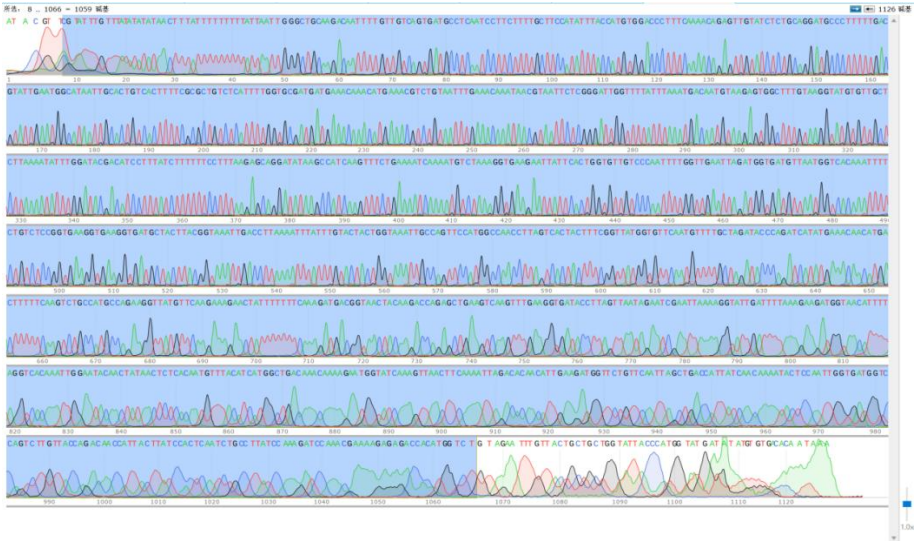
No 15 Date: June 5 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	Yeast broth verification PCR (Protocol 23)
Note	On 5 June the verification PCR was carried out according to the last protocol of the protocol compilation — Protocol 23 (verification PCR from yeast liquid culture, snailase lysis): 1 mL of culture was collected and washed with sorbitol buffer, the cell wall was digested with 100 mg/mL snailase at 37 °C for 1 h, the lysate was heated at 100 °C for 10 min, and 4.6 µL of it was used in the same 10 µL verification mixture and cycling programme.
Result	Samples 1 and 3 gave a clear target band of the expected size (Fig. 2), while the products from samples 2 and 4 were non-specific amplifications.  **Fig. 2 Agarose gel electrophoresis of the verification PCR products from the snailase lysates: samples 1 and 3 show the target band of the expected size, while samples 2 and 4 show non-specific amplification.**
Explanation	The milder snailase treatment releases the template without destroying it, whereas the strong alkali used on 4 June (No. 14) degraded the template and/or inactivated the polymerase. The positive bands of samples 1 and 3 therefore indicate that the construct has been integrated into these transformants.
Next plan	The positive samples 1 and 3 will be sent for Sanger sequencing to confirm the insert sequence and its orientation; if the sequencing result is correct, the positive clones will be expanded for subsequent functional verification.

No 16 Date: June 6 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	Glycerol stock of verified transformants
Note	On June 6, glycerol stocks were prepared from the expanded cultures of the two single colonies that had been verified by PCR (see No. 15). 0.75 mL of each culture was transferred directly into a cryogenic tube, 0.75 mL of 50% (v/v) glycerol was added, and the mixture was vortexed thoroughly. The tubes were placed at -20 °C for 30 min and then transferred to a -80 °C freezer for long-term storage. Four tubes were prepared from each of the two verified cultures, giving eight tubes in total.
Result	Eight glycerol stocks were prepared (four for each verified clone) and are stored at -80 °C. The stocks have been kept for subsequent reactivation, plasmid extraction and functional verification, and no loss or contamination was observed during this step.
Explanation	Glycerol at a final concentration of about 25% (v/v) serves as a cryoprotectant: it lowers the freezing point and suppresses the formation of intracellular ice crystals, thereby reducing mechanical damage to the cells. The two-step cooling scheme, i.e. pre-equilibration at -20 °C for 30 min before transfer to -80 °C, lessens thermal shock and improves cell viability. Because the starting material was of relatively low activity (see No. 13), several independent aliquots were prepared for each clone to avoid losing the strain.
Next plan	Functional verification will be carried out after the PCR products have been confirmed by sequencing.

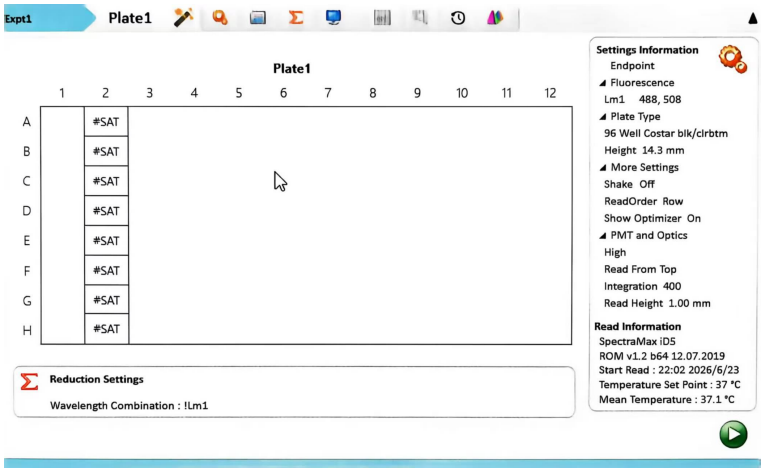
No 17 Date: June 11-June 16 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	Sequencing of verification PCR products
Note	From June 11 to June 16, the verification-PCR products were sent for Sanger sequencing, and the returned reads were aligned against the P_FUS1+yEGFP reference sequence (see Fig. 1).
Result	Sequence alignment shows that the internal main body of the amplified fragment (the sequence highlighted in blue) is identical to the P_FUS1+yEGFP reference sequence, with no point mutations or frameshift mutations introduced. Only a small number of differences occur at the two ends of the sequence, which are mainly due to the limited sequencing read length in this direction and the reduced base-call signal quality toward the termini. These terminal differences do not affect the reading frame or the coding-sequence integrity of the reporter gene, confirming the sequence correctness of the constructed fluorescent reporter system.  **Fig. 1 Sequencing alignment of the PCR product against the P_FUS1+yEGFP reference: the internal region matches the reference exactly, with only terminal base-call noise.**
Explanation	The limited read length of a single Sanger reaction and the decay of the fluorescence signal near the ends of the read account for the few mismatches at the sequence termini. These differences lie in the primer/flanking region rather than the coding region, so they do not alter the open reading frame or the encoded protein of yEGFP.
Next plan	With the sequence confirmed, the verified clone (see No. 15) will proceed to functional verification of the fluorescent reporter system; the corresponding glycerol stocks (see No. 16) can be reactivated as needed.

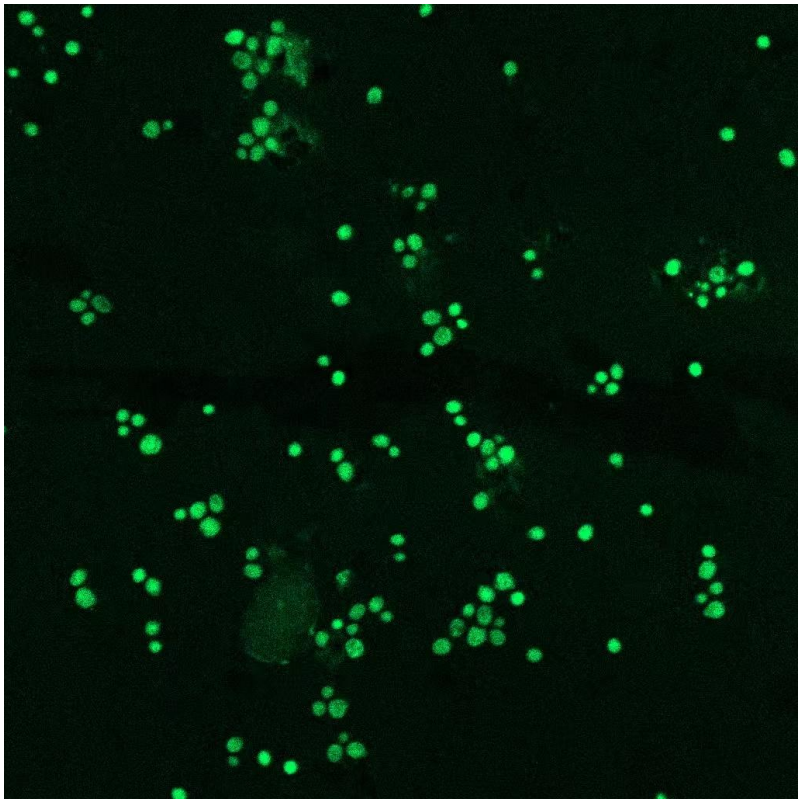
No 18 Date: June 21-June 23 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	Expansion of verified transformants
Note	10 μL of the prepared glycerol stock (see No. 16) was inoculated into 5 mL SD-HIS liquid medium and cultured at 30 °C with shaking at 250 rpm, from 15:55 on 21 June to 11:20 on 23 June; the OD600 of the two cultures reached 2.2566 and 2.0407, respectively.
Result	The glycerol stocks revived and grew well in the selective SD-HIS liquid medium, reaching a high cell density (OD600 = 2.2566 and 2.0407), which provided sufficient biomass for the subsequent induction assay.
Explanation	The HIS auxotroph was maintained under SD-HIS selective pressure, which favours retention of the plasmid; the two cultures reached comparable and sufficiently high OD600 values, indicating a successful expansion.
Next plan	The expanded cultures will be diluted to a starting OD600 of about 0.2 and subjected to α-factor gradient induction and plate-reader fluorescence / OD600 detection (see No. 19).

No 19 Date: June 23 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	α-factor induction and *yEGFP* plate-reader detection
Note	On 23 June each expanded culture (see No. 18) was diluted into four tubes to a starting OD600 of about 0.2; after roughly 3 h of growth the OD600 of the eight tubes were 0.4587, 0.4449, 0.3815, 0.4371, 0.3980, 0.3850, 0.4022 and 0.3154, respectively. Each set of four tubes was then induced with a final-concentration gradient of α-factor (120 µM, 12 µM, 1.2 µM and 0 µM), and plate-reader fluorescence and OD600 were measured at intervals.
Result	Because of inexperience with the plate reader, no usable data were obtained — the fluorescence reading wells all returned a signal-saturation flag (#SAT), as shown in Fig. 1.  **Fig. 1 Plate-reader result showing signal saturation (#SAT) in the reading wells**
Explanation	The #SAT flag means the fluorescence signal exceeded the instrument's upper detection limit (a saturated read) rather than a true endpoint; it reflects a setup/method problem caused by inexperience with the plate reader (e.g. gain / gain-range, read mode or sample dilution) rather than a biological result.
Next plan	After becoming familiar with plate-reader operation, the α-factor induction together with fluorescence and OD600 measurements will be repeated using appropriate gain and dilution settings.

No 20 Date: June 30 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	Confocal microscopy of yEGFP expression
Note	On 30 June, 100 μL of the induced culture that had been stored at 4 °C (see No. 19) was centrifuged and the supernatant discarded. A drop of distilled water was placed on a glass slide, the cells were spread evenly with an inoculation loop, and the slide was briefly heated over an alcohol lamp to remove excess water; when only a small amount of moisture remained, a coverslip was applied. The sample was then observed under a confocal fluorescence microscope using a 40× objective with 488 nm excitation (see Fig. 1).
Result	A large number of yeast cells emitting green fluorescence were observed under the microscope (Fig. 1), preliminarily demonstrating that fluorescent induced expression was successful.  **Fig. 1 Confocal fluorescence image of yeast after α-factor induction (40× objective, 488 nm excitation), showing green fluorescent cells.**
Explanation	The green fluorescence observed under 488 nm excitation matches the emission characteristics of yEGFP, indicating that after α-factor induction the P_FUS1 promoter successfully drove expression of the yEGFP reporter gene. The cells retained an intact morphology and the fluorescence was evenly distributed.
Next plan	Negative controls (e.g. an uninduced wild-type strain or empty-vector transformants) should be included in subsequent experiments to rule out cellular autofluorescence and more fully demonstrate that the fluorescence signal originates from induced expression.

No 21 Date: July 7 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	Vector and fragment amplification for assembly
Note	On July 7, three PCR reactions were set up with a high-fidelity polymerase mix for construct assembly. (1) Linear amplification of the pESC vector backbone — template: 2 μL of pESC-His-yEGFP plasmid (sample 1, 49.850 ng/μL; sample 2, 39.950 ng/μL); primers 1.5 μL each (10 μM); nuclease-free water 20 μL; high-fidelity mix 25 μL. Cycling: 98 °C 5 min; (98 °C 10 s, 55 °C 15 s, 72 °C 4 min) × 30; 72 °C 7 min; 4 °C hold. (2) *FUS1* fragment with homology arms — template: 5 μL of the *FUS1* PCR product diluted 10-fold (sample 1, 8.9 ng/μL; sample 2, 10.2 ng/μL); primers 1 μL each (10 μM); nuclease-free water 18 μL; high-fidelity mix 25 μL. Cycling: 98 °C 3 min; (98 °C 10 s, 55 °C 15 s, 72 °C 15 s) × 30; 72 °C 5 min; 4 °C hold. (3) *lacZα* fragment with homology arms — template: 1 μL of pRS415 plasmid (sample 1, 138.45 ng/μL; sample 2, 112.50 ng/μL); primers 1.5 μL each (10 μM); nuclease-free water 21 μL; high-fidelity mix 25 μL. Cycling: 98 °C 3 min; (98 °C 10 s, 55 °C 15 s, 72 °C 20 s) × 30; 72 °C 5 min; 4 °C hold.
Result	All three PCR reactions were completed on the thermocycler, yielding the linearly amplified pESC backbone together with the homology-arm *FUS1* and *lacZα* fragments; the amplification products were stored at −20 °C for downstream assembly. Agarose-gel electrophoresis verification of the products will be recorded separately.
Explanation	PCR was used to linearize the pESC plasmid as the vector for homology-based assembly, while the *FUS1* and *lacZα* fragments were amplified with primers carrying homologous ends so that they can be joined seamlessly to the linearized backbone. Diluting the *FUS1* template and adjusting the template volume kept the template concentration within the optimal range for the high-fidelity polymerase, and the three reactions shared a 55 °C annealing temperature.
Next plan	The amplified products will be verified by agarose-gel electrophoresis, purified and then used for homology-recombination (Gibson) assembly of the target construct.

No 22 Date: July 8 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	Gel check, purification, Gibson cloning and PCR verification
Note	Steps refer to Protocol 5 (Agarose Gel Electrophoresis), Protocol 10 (Gel Extraction of PCR Products), Protocol 11 (Gibson Assembly) and Protocol 7 (Conventional PCR, standard Taq) in the protocol. Key params: on 8 July the previous-day PCR products (see No. 21) were identified by 1% agarose gel stained with GelRed and run at 80 V (11:15-12:35); each lane 5 μL product + 1 μL 6x Loading Buffer, with 15000 bp and 2000 bp markers on the two sides (Fig. 1). The correct bands were excised and recovered by gel extraction; the linearized pESC backbone with the *FUS1* and *lacZα* homology-arm fragments was then assembled by Gibson one-step cloning at 50 C for 30 min. On 8 July the Gibson assembly products were used as template for PCR verification with a Mix containing green dye (94 C 3 min; 94 C 30 s, 55 C 30 s, 72 C 30 s x30; 72 C 5 min), and the products were checked by 1% agarose gel electrophoresis with a 2 kb marker at 80 V (20:52-21:49) (Fig. 2). Parameter differences: electrophoresis run at 80 V for the times above (Protocol 5 default about 90 V, 50 min); the Gibson reaction extended to 50 C for 30 min (Protocol 11 default 50 C, 15 min); the verification PCR used a green-dye-containing Mix (Protocol 7).
Result	The electrophoresis of the previous day's PCR products (Fig. 1) showed a single correct band for each reaction — the linearised pESC backbone and the *FUS1* and *lacZα* homology-arm fragments all migrated to the expected sizes, confirming that the PCRs were specific and suitable for downstream cloning. In the verification PCR of the Gibson assembly products (Fig. 2), a band of the correct size was observed, but many additional non-specific bands were also present.



Fig. 1 Agarose gel electrophoresis of the PCR products amplified the previous day (15,000 bp and 2,000 bp markers).

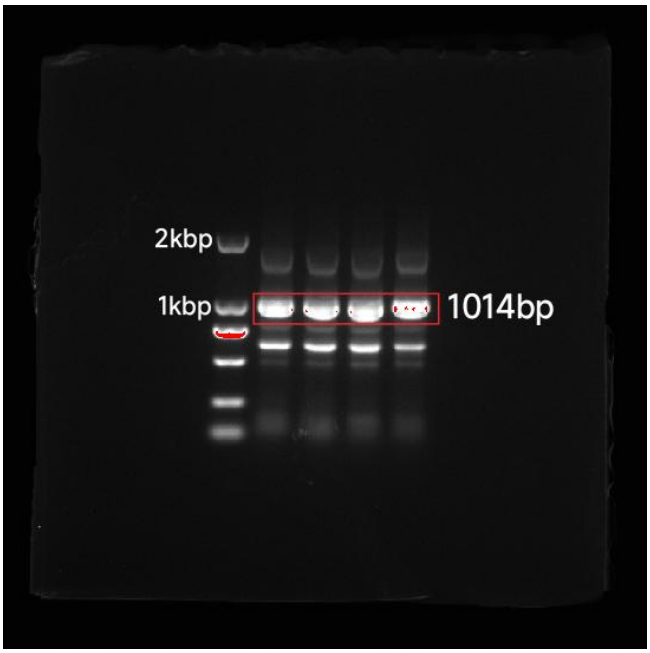
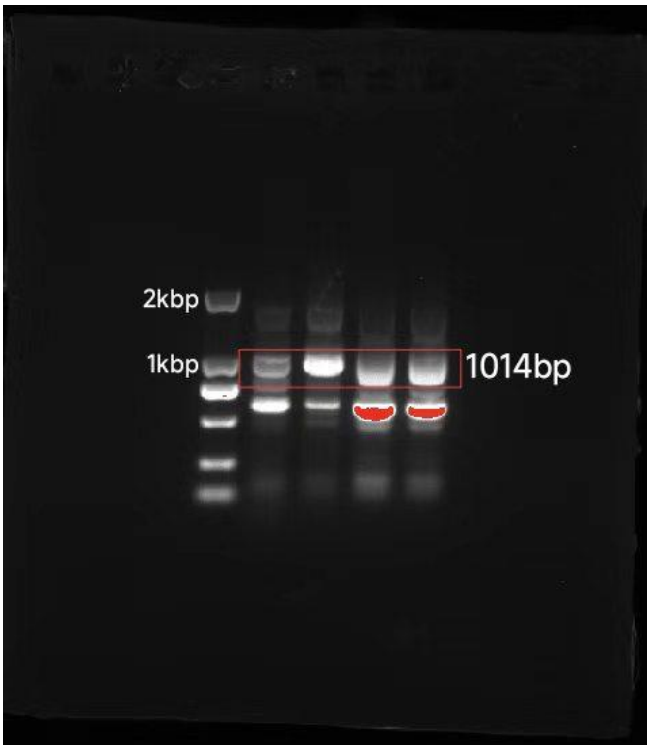


Fig. 2 Agarose gel electrophoresis of the verification-PCR products from the Gibson assembly (2 kb marker).

Explanation	The clean single bands in Fig. 1 indicate specific amplification of both the linearised backbone and the homology-arm fragments. The extra bands in Fig. 2 suggest that the Gibson one-step cloning system still suffers from non-specific amplification and insufficient assembly specificity, most likely related to the

	insert-to-vector ratio, the primer annealing and extension conditions, or the fidelity of the system used.
Next plan	The Gibson one-step cloning system will be further optimised (e.g. by adjusting the insert-to-vector molar ratio, the annealing/extension conditions or the assembly system) to improve the specificity and efficiency of assembly, after which the products will be transformed and confirmed by sequencing.

No 23 Date: July 9 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	Gibson cloning optimization and gel verification
Note	On 9 July the Gibson one-step assembly was repeated with the optimised system. The fragments gel-purified the previous day (PCR-purified samples 1, 2, 4 and 5; A260/A280 1.44–1.52, concentration 131.8–193.0 ng/μL, some diluted with 2 μL sample + 3 μL sterile water) were joined with 2× CE Mix at 50 °C for 30 min and held at 4 °C. Each assembly product (1–4) was then used as template for verification PCR (Mix 2.5 μL, sterile water 20 μL, primers 2 μL each, template 1 μL; 94 °C 3 min; 94 °C 30 s, 55 °C 30 s, 72 °C 1 min ×30 cycles; 72 °C 5 min; 4 °C overnight), and the products were resolved on a 1% agarose gel (GelRed, 80 V, 12:18–13:18, 2 kb Marker, 5 μL per lane) for inspection (see Fig. 1).
Result	Agarose gel electrophoresis of the verification PCR products (Fig. 1) showed that, although bands of the expected size were present, many non-specific impurity bands co-existed, so a single correct band could not be obtained even after the Gibson one-step cloning system had been optimised.  **Fig. 1 Agarose gel electrophoresis of the verification PCR products of the optimised Gibson assembly (2 kb Marker).**
Explanation	Although bands of the expected size were present, multiple non-specific bands remained in the verification PCR even after the system had been optimised, indicating that the ligation specificity of this one-step Gibson system could not be improved to yield a single correct amplicon; further tweaking of fragment ratios and conditions would be time-consuming with uncertain benefit.

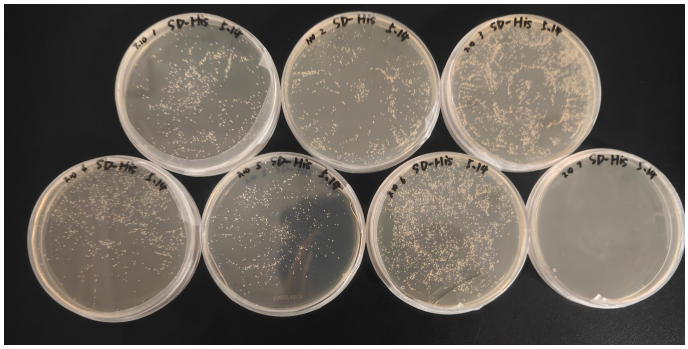
Next plan	Rather than continuing to purify a single correct band by gel, the verification PCR products (which contain the correct band within the mixture) will be transformed directly into competent cells, relying on subsequent antibiotic / blue-white screening to enrich positive clones, then confirmed by colony PCR and sequencing.

No 24 Date: July 10 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	BY4741 competent-cell preparation
Note	On 10 July BY4741 was revived from a single colony picked from a non-selective plate, grown overnight, then inoculated (9:56) into two 50 mL YPD liquid medium flasks (labelled 1 and 2) and incubated at 30 °C, 200 rpm. Growth was monitored by OD600: the initial OD600 was about 0.19–0.24; at 11:23 the two flasks read 0.2910 and 0.2801, were returned to shaking (11:33), and at 12:55 read 0.4761 and 0.4622 — within the 0.4–0.6 exponential window. The two cultures were labelled GS1 and GS2, and competent cells were prepared as follows, referring to the competent-cell part of Protocol 12 (Yeast Transformation, incl. competent cell preparation): centrifuge at 3000 × g for 3 min at room temperature and discard the supernatant; resuspend the pellet in 7 mL Buffer A and re-centrifuge at 3000 × g for 3 min; resuspend in 400 µL Buffer B and incubate at room temperature for 10 min. Parameter differences vs Protocol 12: (i) the culture volume was 50 mL per flask (Protocol 12 is based on 10 mL of culture); (ii) GS1 was aliquoted as 3 × 100 µL and GS2 as 4 × 150 µL into 1.5 mL sterile tubes (tubes labelled 1–7), whereas Protocol 12 aliquots 100 µL per tube.
Result	Competent cells were successfully prepared from both cultures (GS1 and GS2) and aliquoted into 1.5 mL sterile centrifuge tubes — tubes 1–3 from GS1 (100 µL each) and tubes 4–7 from GS2 (150 µL each). The aliquots were kept chilled and are ready for the subsequent yeast transformation.
Explanation	The OD600 of both cultures reached 0.4–0.6 by 12:55, the exponential phase required for efficient competent-cell preparation. The Buffer A wash removes the culture medium and Buffer B renders the cells permeable and competent, consistent with the competent-cell procedure of Protocol 12.
Next plan	Use the prepared competent cells for the subsequent yeast transformation, and store a portion at −80 °C as backup stocks.

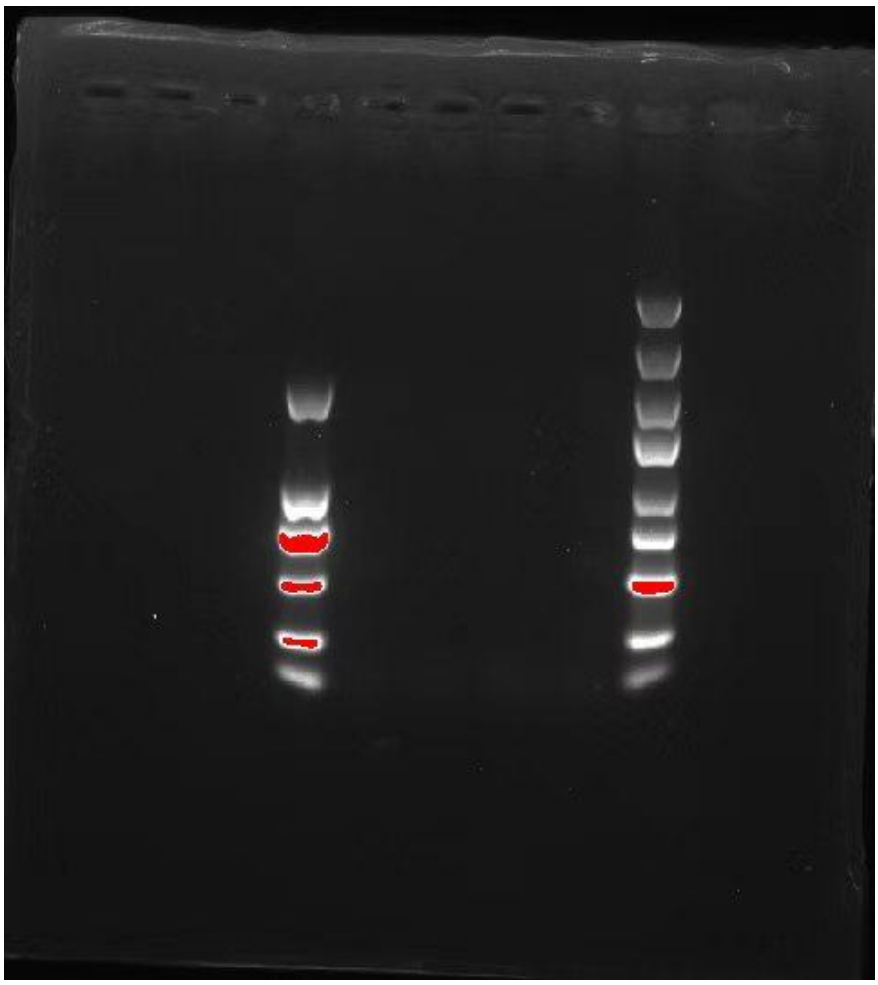
No 25 Date: July 10-July 12 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	Yeast transformation and SD-His selection
Note	On 10 July, using the BY4741 competent cells prepared the same day (see No. 24), the Gibson assembly products obtained on 8 and 9 July (see No. 22 and No. 23) were transformed into yeast. 80 µL of Carrier DNA was aliquoted into a 15 mL sterile centrifuge tube, heat-denatured in a 100 °C metal bath for 15 min and immediately placed on ice. Each 5 µL of an assembly product was mixed with 10 µL of the heat-denatured Carrier DNA by pipetting in a 2 mL sterile tube, then added to a competent-cell sample: 7.8 product 1 → sample 1, 7.9 product 1 → sample 2, 7.9 product 3 → sample 3, 7.8 product 2 → sample 4, 7.9 product 2 → sample 5, 7.9 product 4 → sample 6; 5 µL of sterile nuclease-free water was added to sample 7 as the negative control. Transformation steps refer to Protocol 12 (Yeast Transformation): the seven samples were transferred into 600 µL of Buffer B, mixed by inversion and incubated in a 30 °C metal bath for 30 min; 70 µL of DMSO was then added, mixed gently by inversion, incubated in a 42 °C metal bath for 15 min and immediately placed on ice for 5 min. After centrifugation at 13,000 × g for 10 s the supernatant was discarded, the cells were resuspended in 100 µL of Buffer and spread evenly onto SD-His plates. The plates were labelled 1–7 and incubated inverted at 30 °C (started at 15:45), then removed at 19:13 on 12 July, sealed with parafilm, photographed and stored at 4 °C. Key parameters / differences from Protocol 12: (i) 5 µL of crude Gibson product was used directly as the template instead of quantified plasmid DNA (0.1–1 µg in Protocol 12); (ii) Buffer B was used (Protocol 12 step b specifies Buffer C); (iii) incubation in a 30 °C metal bath for 30 min (Protocol 12: water bath 45 min); (iv) the GS1/GS2 competent cells prepared in No. 24 were used, whose aliquot volumes differ from the 100 µL per tube in Protocol 12; (v) Carrier DNA was heat-denatured freshly at 100 °C for 15 min before use.
Result	The transformation results are shown in Fig. 1. On the SD-His (histidine-deficient) selective plates, plates 1–6 showed colonies of varying density (plates 3 and 6 nearly covered the whole surface, the others sparser), indicating that HIS^+ transformants carrying the recombinant plasmid grew successfully; plate 7, the sterile-water negative control, showed no colony growth.

	 **Fig. 1 Colonies on SD-His selective plates after yeast transformation (plate 7 = sterile-water negative control, no growth).**
Explanation	The SD-His medium lacks histidine, so only transformants that have taken up the recombinant plasmid (complementing the HIS3 marker) can grow, enabling positive selection. The complete absence of colonies on plate 7 (sterile water) confirms that there was no contamination or false-positive growth on the selective plates, supporting that the colonies on plates 1–6 arose from genuine transformation.
Next plan	Pick single colonies from plates 1–6, expand them in liquid medium, and verify the correct insertion of the *FUS1*/*lacZα* constructs by colony PCR (Protocol 13) and sequencing.

No 26 Date: July 13 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	Liquid culture and colony PCR of *lacZα* transformants
Note	On 13 July, single colonies were picked with a sterile white tip from different areas of the *lacZα*-transformant SD-His plate No. 5 (see No. 25) - 4 colonies of varying size - and inoculated into four sterile tubes each containing about 5 mL of SD-His liquid medium, labelled (1)(2)(3)(4), then cultured in a shaking incubator at 30 °C and 250 rpm (started at 19:48). The liquid expansion follows No. 13. Colony-PCR verification was performed the same day (steps refer to Protocol 13, S. cerevisiae colony-PCR). Four single colonies were picked from different areas of plate No. 5, each suspended in 20 µL of sterile water in a sterile 0.2 mL tube and mixed; 10 µL of each was transferred to a new empty 0.2 mL tube, labelled 1, 2, 3, 4 (matching (1)(2)(3)(4)). The tubes were lysed by the transformation-verification method of Protocol 13: suspended in sterile water and cycled at 98 °C for 3 min and 4 °C for 2 min for 5 cycles, then heated at 98 °C for 3 min and stored at 4 °C. Parameter difference: Protocol 13 also lists a mild 20 mM NaOH lysis, whereas the sterile-water thermal lysis of its transformation-verification method (98 °C 3 min, 4 °C 2 min, 5 cycles) was used here. PCR mixture (50 µL each): Mix (with green loading dye) 25 µL, sterile water 13 µL, forward and reverse primers (100 µM) 2 µL each, template (the lysed samples above) 8 µL. Cycling: 94 °C 5 min; 94 °C 30 s, 55 °C 30 s, 72 °C 1 min for 30 cycles; 72 °C 5 min; 16 °C hold. Parameter differences: a green-dye pre-mixed Mix was used instead of standard Taq, with 55 °C annealing for 30 cycles. Electrophoresis (steps refer to Protocol 5, agarose gel electrophoresis): 1% agarose gel stained with GelRed, run at 80 V (18:06-18:07). Samples loaded: 2 kb Marker, samples 1-4 and 5 kb Marker, 5 µL each. Parameter difference: the protocol defaults to about 90 V for 50 min, whereas 80 V with a short run time was used here, and both 2 kb and 5 kb markers were applied.
Result	The electrophoresis result is shown in Fig. 1. No band of the expected size was detected in lanes 1-4, whereas the 2 kb / 5 kb markers showed clear bands.

	 **Fig. 1 Agarose gel electrophoresis of the colony-PCR products (no visible bands in lanes 1-4).**
Explanation	The absence of bands is most likely because the picked single colonies were too small, giving insufficient cell mass, or the lysis was incomplete, so the template DNA did not reach an amplifiable concentration; mismatched primer/annealing conditions may also contribute. The liquid cultures set up in parallel are still growing and can be used to repeat the test.
Next plan	Re-pick larger single colonies with more cell mass and strengthen/extend the lysis (or switch to the NaOH lysis of Protocol 13) before repeating the colony PCR and electrophoresis; the liquid cultures can also be used to enrich biomass for verification, and positive clones will be confirmed by sequencing of the *FUS1*/*lacZα* insert.

No 27 Date: July 14 Experimenters: Chengxi Luo Recorder: Chengxi Luo

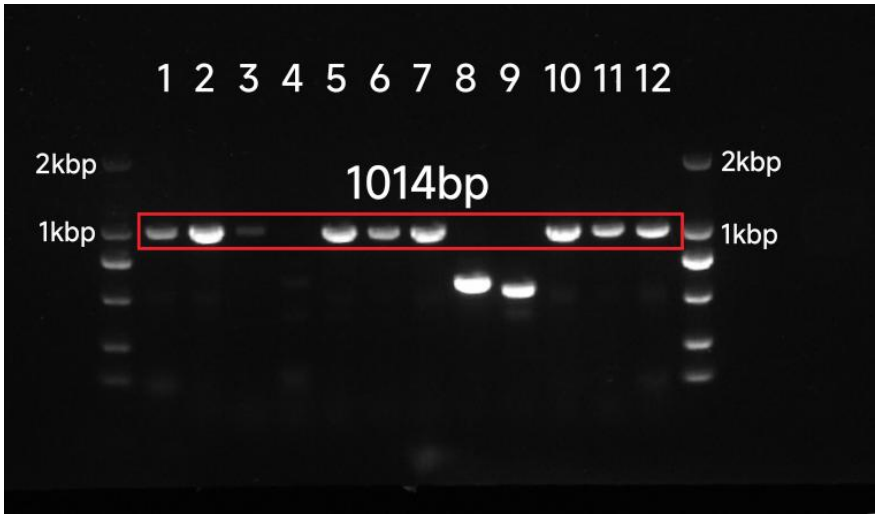
Experiment	Broth PCR verification and template optimization
Note	On 14 July the previous day's transformant liquid cultures (see No. 26) were used for broth-colony (bacterial-lysate) PCR verification. At 15:12, 1.5 mL was taken from each tube into a sterile 1.5 mL microcentrifuge tube and labelled (1)(2)(3)(4); at 15:21 the culture tubes were returned to the shaker at 30 °C and 250 rpm for further growth. The microcentrifuge tubes were centrifuged at 12000 rpm at 4 °C for 2 min and the supernatant discarded; 200 µL of sorbitol (lysis) buffer and 20 µL of 100 mg/mL snailase were added to each, vortexed and incubated at 37 °C for 1 h to break the cell wall. Cells were then pelleted at 3000 rcf, 4 °C for 5 min and the supernatant removed; 50 µL sterile water was added to tube (1) and 20 µL to the others, vortexed, and lysed at 85 °C for 10 min. PCR verification (steps refer to Protocol 23, broth-colony PCR). The 50 µL mixture was: Mix (with green loading dye) 25 µL, sterile water 20 µL (15 µL for sample 4), forward and reverse primers (10 µM) 2 µL each, template (the lysed samples above) 1 µL (6 µL for sample 4). Cycling: 94 °C 3 min; 94 °C 30 s, 55 °C 30 s, 72 °C 1 min 10 s for 28 cycles; 72 °C 5 min; 16 °C hold. Parameter difference: a green-dye pre-mixed Mix was used, with 10 µM primers, 55 °C annealing, 1 min 10 s extension and 28 cycles. Electrophoresis (steps refer to Protocol 5, agarose gel electrophoresis): 1% agarose gel stained with SYBR Safe, run at 120 V (18:39-19:10). Loading: 2 kb Marker and PCR samples 1-4, 5 µL each. Samples 1-3 gave faint bands of the expected size with primer-dimer at the gel bottom, and sample 4 showed a faint non-specific band. Parameter difference: the protocol defaults to about 90 V for 50 min, whereas 120 V with a short run was used here. Because the target bands were faint, the template amount was increased and the PCR repeated: Mix 25 µL, sterile water 15 µL, primers (10 µM) 2 µL each, template 6 µL, with the cycling conditions as above but the number of cycles raised to 30. A second gel (1% agarose, SYBR Safe, 120 V, about 2 h, 19:26-21:56; 2 kb Marker and PCR samples 1-4, 5 µL each) was run.
Result	The first electrophoresis is shown in Fig. 1: samples 1-3 gave visible target bands of the expected size but faint, with primer-dimer at the bottom, and sample 4 showed a faint non-specific band. After increasing the template amount, the second electrophoresis (Fig. 2) gave stronger target bands; the band in lane 2 was caused by overflow (cross-contamination) of the sample from lane 1.



Fig. 1 Agarose gel electrophoresis of the first broth-colony PCR (faint target bands in lanes 1-3).

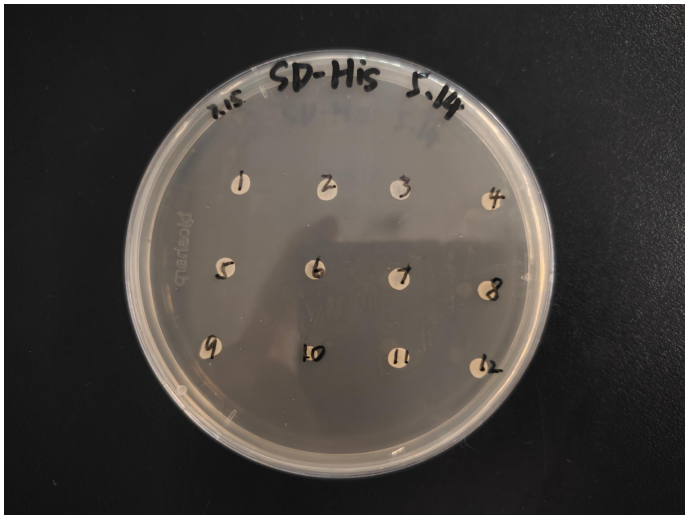
	 **Fig. 2 Electrophoresis after template-volume optimization (stronger target bands; lane 2 due to overflow from lane 1).**
Explanation	The faint initial bands were most likely due to the low template amount (1 μL of lysate per tube, insufficient cell mass); increasing the template to 6 μL and extending the cycles strengthened the bands. The extra band in lane 2 arose from overflow of the lane 1 sample during loading, not from amplification of that sample itself.
Next plan	Continue liquid culture of the clones matching the clear bands and confirm the *lacZα*/*FUS1* insert by sequencing; re-load the lane 2 sample independently to re-check it. If bands remain weak, further increase the template amount or adjust the annealing temperature.

No 28 Date: July 15 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	Colony PCR of clones 1-12 and replica plating
Note	On 15 July, 12 single colonies were picked from different areas of the *lacZα*-translation No. 5 SD-His plate (see No. 25) with a white tip and suspended in 10 μL sterile water in 0.2 mL tubes. A fresh SD-His plate was marked in 3 rows × 4 columns (Nos. 1-12) and 2.5 μL of each broth was replica-spotted onto the corresponding position; the plate was then incubated inverted at 30 °C from 10:25. Colony lysis (refer to Protocol 13, transformation-verification method): the tubes were lysed at 98 °C for 3 min and then 4 °C for 2 min for 5 cycles, then centrifuged. Difference: Protocol 13 also lists a mild 20 mM NaOH lysis; here the sterile-water thermal-lysis programme of its transformation-verification method was used, with the leftover broth after replica plating as template. Broth-colony PCR (refer to Protocol 23, broth-colony PCR). The reaction mixture was: Mix (with green loading dye) 5 μL, up- and down-stream primers (10 μM) 0.2 μL each, and 4.5 μL of the corresponding lysed broth. Cycling: 94 °C 3 min; (94 °C 30 s / 55 °C 30 s / 72 °C 1 min 10 s) × 35; 72 °C 5 min; 16 °C hold. Difference: a green-dye pre-mix and 10 μM primers were used, with annealing at 55 °C, extension 1 min 10 s and 35 cycles. Electrophoresis (refer to Protocol 5): 1% agarose gel with GelRed, 80 V, 14:19-15:12; 2 kb Marker, samples 1-12 and 2 kb Marker, 5 μL each. Difference: 1% GelRed gel run at 80 V for a short time (protocol default about 90 V, 50 min).
Result	The electrophoresis result is shown in Fig. 1. Except for samples 4, 8 and 9, all other samples (1-3, 5-7, 10-12) showed a single clear target band of the expected size, though some of them also had a few faint non-specific bands. Samples 4, 8 and 9 gave no target band.  Fig. 1 Agarose gel electrophoresis of the colony PCR products (Nos. 1-12; single clear target bands except Nos. 4, 8 and 9, some with faint extra bands)
Explanation	A single clear target band indicates correct amplification of the target insert (*FUS1*/*lacZα*) from that clone. The few faint extra bands are non-specific

	products caused by slightly excess template or non-specific primer binding. The absence of a band in samples 4, 8 and 9 suggests too little template or inefficient lysis.
Next plan	The clones matching the clear single bands will be picked from the replica plate, expanded and sequenced to confirm the *FUS1*/*lacZα* insert. Samples 4, 8 and 9 will be re-tested by re-picking larger colonies and enhancing lysis. The faint extra bands can be reduced by lowering the template volume or raising the annealing temperature.

No 29 Date: July 16 Experimenters: Chengxi Luo Recorder: Chengxi Luo

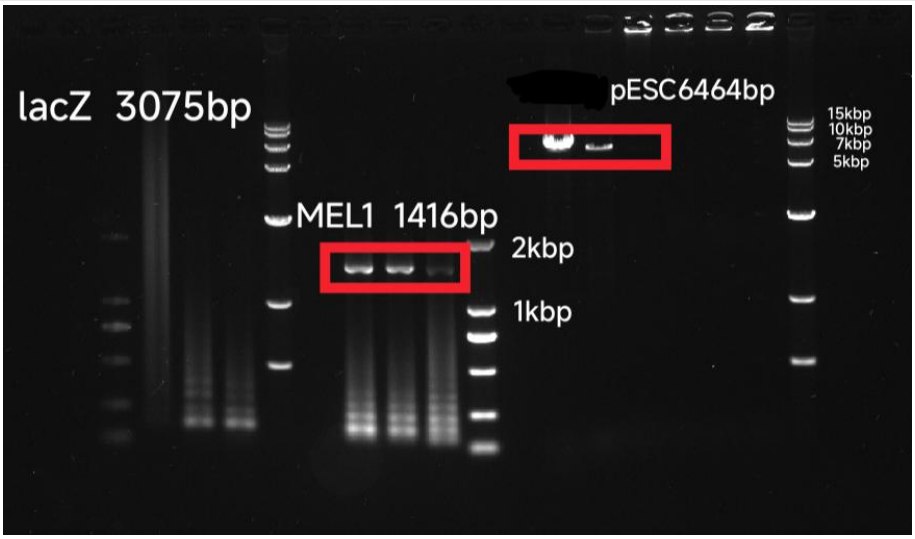
Experiment	α-factor induction and X-gal colour test
Note	On 16 July, clones that had passed colony PCR verification (see No. 28) were subjected to α-factor induction, then incubated in the dark on medium containing the X-gal substrate to detect β-galactosidase colour development.
Result	The result is shown in Fig. 1. After induction, the colonies failed to turn the X-gal substrate and no blue/white colonies were formed.  **Fig. 1 X-gal colour development assay on the SD-His plate after α-factor induction**
Explanation	This indicates that the verified clones still cannot produce functional β-galactosidase, and blue-white colour development has not been established in this system.
Next plan	The system will be redesigned to establish blue-white colour development.

No 30 Date: July 17 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	Revival of fluorescent yeast and AH109 glycerol stocks
Note	On 17 July, 10 μL each of the constructed fluorescent yeast glycerol stock and the AH109 yeast glycerol stock were spotted onto SD-His plates and YPD plates respectively, then streaked and incubated inverted at 30 °C. This revival was carried out to obtain actively growing cells as the source for the subsequent amplification of the gene fragments required to construct the *P_FUS1+MEL1*/*lacZ* reporter system.
Result	Both glycerol stocks were streaked onto SD-His and YPD plates and incubated inverted at 30 °C; lawn/colony growth is to be assessed after incubation.
Explanation	SD-His plates select for yeast retaining the pRS415-based plasmid, while YPD plates support non-selective growth and expansion of the glycerol stocks. Successful revival provides sufficient high-quality genomic template for the subsequent amplification of the *P_FUS1* and *MEL1* fragments and the *lacZ* reporter fragment.
Next plan	Once growth is confirmed, cells will be harvested for genomic DNA preparation and the gene fragments required for the *P_FUS1+MEL1*/*lacZ* reporter system will be amplified; the amplified fragments will then be used to assemble the reporter construct.

No 31 Date: July 20 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	Colony PCR, gel analysis and glycerol-stock activation
Note	On 20 July single colonies were picked with a white tip from the plates streaked on 17 July (see No. 30) - the YPD plate carried single colonies while the SD-His plate had none - and suspended in 12 µL of 20 mM NaOH: colonies from the YPD plate were labelled 1, 2, 3 and three random colonies from the SD-His plate were labelled 1, 2, 3 in circles. The tubes were lysed in a thermocycler at 98 C for 3 min then 4 C for 2 min for 5 cycles, then held at 16 C; after centrifugation the supernatant was used as PCR template. Colony lysis refers to Protocol 13 (transformation-verification method); this time the 20 mM NaOH thermal-lysis programme (98 C 3 min / 4 C 2 min) was used, which differs from the sterile-water lysis used in No. 28. Colony PCR refers to Protocol 13. Each reaction (50 µL): 2x high-fidelity Mix 25 µL, sterile water 18 µL, up- and down-stream primers (10 mM) 1 µL each, and 5 µL of the corresponding lysed-colony template. Three primer pairs were run in parallel - *lacZ*-F/R (templates 1-3), *MEL1*-F/R (templates 1-3) and pESC-F/R (templates 1-3 circled). Cycling: 98 C 3 min; (98 C 15 s / 55 C 15 s / 72 C 3 min, 6 min for the pESC set) x 35; 72 C 5 min; 16 C hold. Electrophoresis refers to Protocol 5: 1% agarose gel with GelRed, 100 V, 17:05-17:57; products were loaded at 3 µL + 0.6 µL 6x loading buffer with 15 kb and 2 kb markers. Difference: 1% GelRed gel run at 100 V for a short time (protocol default about 90 V, 50 min). Glycerol-stock activation: glycerol stocks 3 and 4 were thawed; four tubes each containing 5 mL SD-His liquid medium were labelled 3, 4 and the corresponding circled numbers; 40 µL of the fluorescent *yEGFP* yeast glycerol stock was added to tubes 3 and (3) and 40 µL of stock No. 4 (*yEGFP*) to tubes 4 and (4), then incubated in a shaker at 30 C, 200 rpm from 11:02.
Result	The colony-PCR products were resolved by agarose-gel electrophoresis (Fig. 1). The *lacZ*-F/R reactions showed no specific band, with only a bright primer-dimer band near the bottom of the gel; the *MEL1*-F/R reactions gave a single clear band of the expected size (about 1.4 kb, *MEL1* 1416 bp); the pESC-F/R reactions gave a target band at about 6.4 kb (pESC 6464 bp) in only the first two lanes.

	 **Fig. 1 Agarose-gel electrophoresis of the colony-PCR products (*lacZ*-F/R / *MEL1*-F/R / pESC-F/R; 15 kb and 2 kb markers)**
Explanation	The *MEL1* fragment (about 1416 bp) and part of the pESC backbone (about 6464 bp) amplified correctly and can serve as template for the *P_FUS1+MEL1* reporter assembly. The *lacZ*-F/R reaction failed to amplify a product (primer-dimer only).
Next plan	The amplified *MEL1* fragment and the linearized pESC fragment will be purified and used as the template for assembly of the *P_FUS1+MEL1* reporter system, and the *lacZ* fragment PCR program will be optimized for re-amplification.

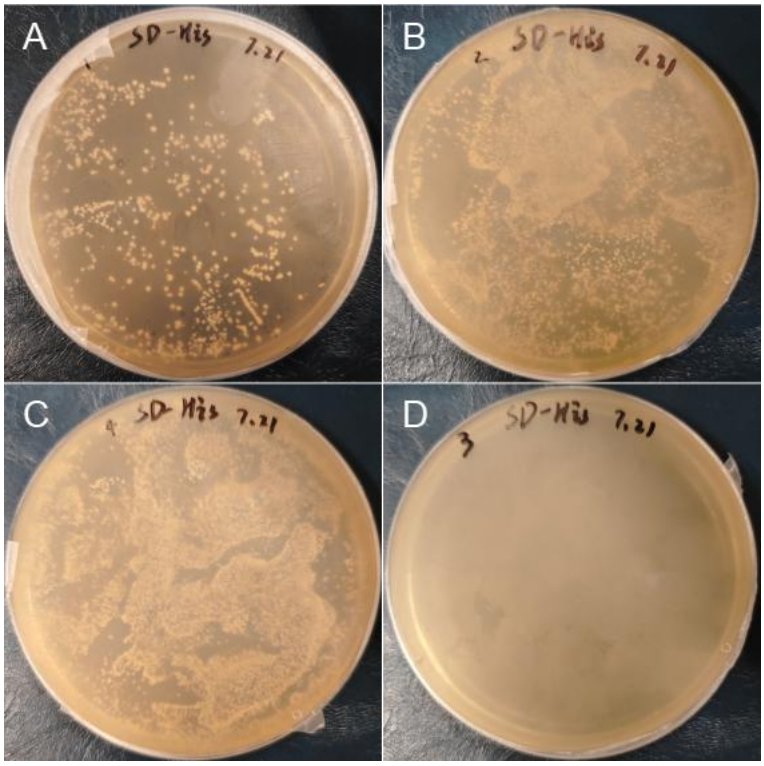
No 32 Date: July 21 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	*lacZ* optimization, *MEL1* gel extraction and pESC purification
Note	Work was carried out in the order: *lacZ*-fragment PCR optimization, agarose-gel electrophoresis, then gel extraction and PCR purification. *lacZ* optimization: to address the failed *lacZ* amplification in No.31, a two-step program was redesigned (same ~25 µL system): 98 C 3 min; (98 C 15 s, 55 C 15 s, 72 C 1 min 40 s) x 10 cycles; (98 C 15 s, 65 C 15 s, 72 C 1 min 40 s) x 20 cycles; 72 C 5 min; hold 16 C. First 10 cycles anneal at 55 C, last 20 at 65 C, each followed by a 72 C 1 min 40 s extension. Electrophoresis refers to Protocol 5 (differences noted): 1% GelRed gel; the first 6 lanes were the previous-day (No.31) *MEL1* PCR products and the last 3 lanes the *lacZ*-optimized products. *MEL1* gel extraction refers to Protocol 10 (differences noted): *MEL1* bands from the first 6 lanes were cut (Buffer QG dissolution, Buffer DW2 wash, EB elution); only two tubes were extracted - *MEL1*-1 = lanes 1, 3, 5; *MEL1*-2 = lanes 2, 5, 6. pESC purification refers to Protocol 11 (PCR clean-up; differences noted): the previous-day pESC-F/R products (about 47 µL + 47 µL) were purified directly (no gel slicing), giving one tube of linearized pESC.
Result	The first 6 lanes (previous-day *MEL1* PCR products) showed a single clear band at about 1.4 kb; the last 3 lanes (*lacZ*-optimized products) showed no band at all. NanoDrop: *MEL1*-1 A260/A280 = 1.898 (20.500 ng/µL), *MEL1*-2 = 2.093 (23.900 ng/µL), linearized pESC = 1.873 (151.40 ng/µL).  **Fig. 1 Agarose-gel electrophoresis of the day**
Explanation	The *MEL1* fragment purified well (A260/A280 about 1.9-2.1) and the linearized pESC was of acceptable purity (about 1.87), both suitable as templates for assembling the *P_FUS1+MEL1* reporter system. After two-step optimization the *lacZ* reaction still gave no band; likely because the full-length *lacZ* sequence on AH109 was unclear and primer design was problematic, so the *P_FUS1+lacZ*

	reporter system was abandoned.
Next plan	The purified products will be transformed into BY4741 yeast, exploiting the yeast's own homologous recombination to construct the *P_FUS1+MEL1* reporter system.

No 33 Date: July 21 Experimenters: Chengxi Luo Recorder: Chengxi Luo

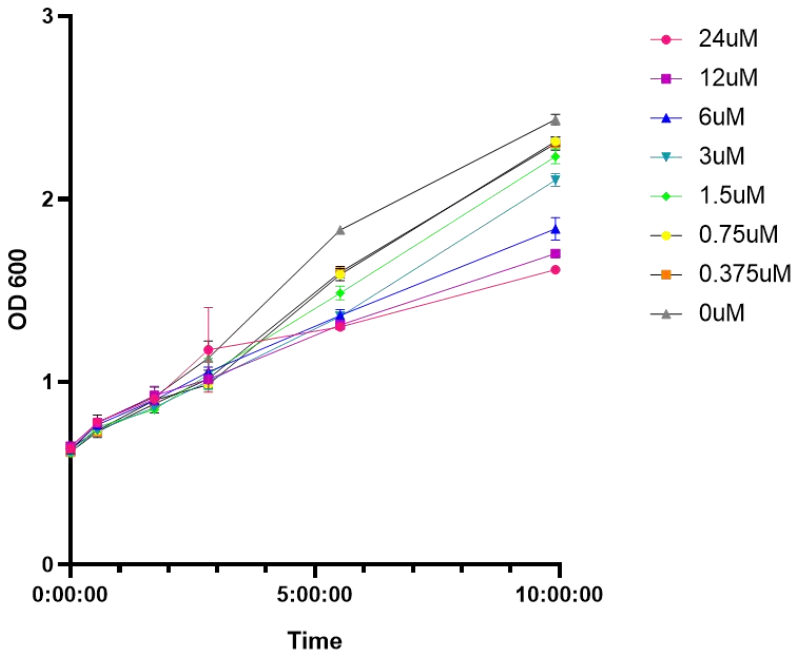
Experiment	Yeast transformation with the *MEL1* and pESC fragments
Note	Today the *MEL1* fragment and the linearized pESC fragment purified in No.32 were co-transformed into competent BY4741 yeast, relying on yeast-mediated homologous recombination to assemble the *P_FUS1+MEL1* reporter system; the procedure refers to Protocol 12 (Yeast Transformation, incl. competent-cell preparation), with only the differing parameters noted. Carrier DNA treatment: 40 μL of Carrier DNA was aliquoted into a 1.5 mL tube, heat-denatured in a 100 C metal bath for 10 min, then placed immediately on ice. Assembly setup (three 0.2 mL tubes): tube 1 = 2 μL linearized pESC (151.40 ng/μL) + 3 μL *MEL1*-1 (20.500 ng/μL); tube 2 = 2 μL linearized pESC (151.40 ng/μL) + 3 μL *MEL1*-2 (23.900 ng/μL); tube 3 = 5 μL sterile water (negative control); 10 μL of heat-denatured Carrier DNA was added to each tube and mixed. 100 μL of competent yeast was added to each tube (labelled 1, 2, 3) and mixed by flickering; 600 μL of Buffer C was added and mixed by inversion, incubated in a 30 C metal bath for 30 min, then 70 μL of DMSO was added and mixed gently by inversion. Heat shock: 15 min in a 42 C metal bath, then immediately 5 min on ice; centrifuged first at 13000 rcf for 10 s (pellet not visible), then at 14000 rcf for 10 s; the supernatant was discarded and the cells were resuspended in 100 μL of Buffer A. Plating (SD-His dropout plates): plates 1 and 4 received 100 μL and 90 μL of transformation sample 1 respectively; plates 2 and 3 each received 100 μL of transformation samples 2 and 3; incubated inverted at 30 C (started at 19:57).
Result	The transformation was completed as planned and the four SD-His dropout plates were incubated inverted at 30 C (from 19:57 on July 21 to 12:26 on July 24, when the culture was ended and the result was read). After further incubation, as shown in Fig. 1, positive transformation plates A, B and C all grew colonies (of uneven density), while the negative-control plate D showed no colony growth, consistent with the expectation and indicating that the transformation worked and the plates were not contaminated.

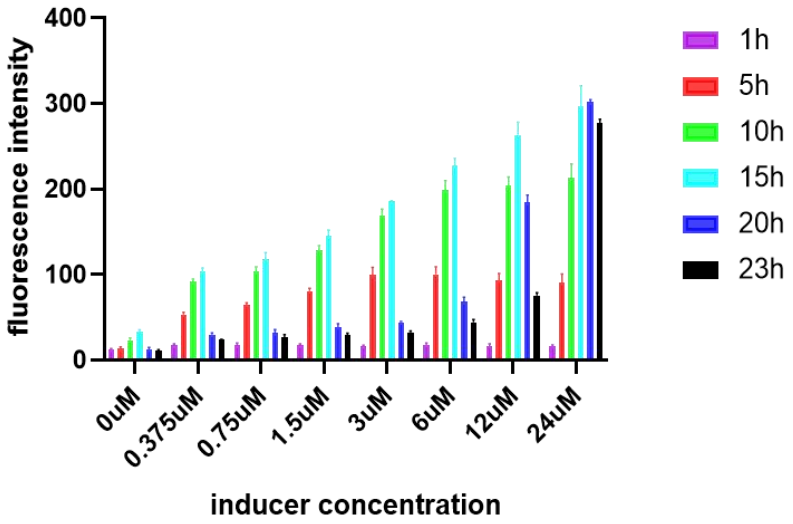
	 **Fig. 1 Colony growth on the SD-His transformation plates**
Explanation	The linearized pESC vector and the *MEL1* fragment were co-transformed, and the *P_FUS1+MEL1* reporter system is expected to be assembled in vivo by yeast homologous recombination; transformation sample 3 (5 μL sterile water) served as the negative control to rule out contamination of the competent cells and the plates, and no colonies are expected on it (negative plate D in Fig. 1).
Next plan	After the transformants grow into single colonies, colonies will be picked for colony PCR verification (referring to Protocol 13) to confirm whether the *P_FUS1+MEL1* reporter system has been successfully constructed.

No 34 Date: July 22 Experimenters: Chengxi Luo Recorder: Chengxi Luo

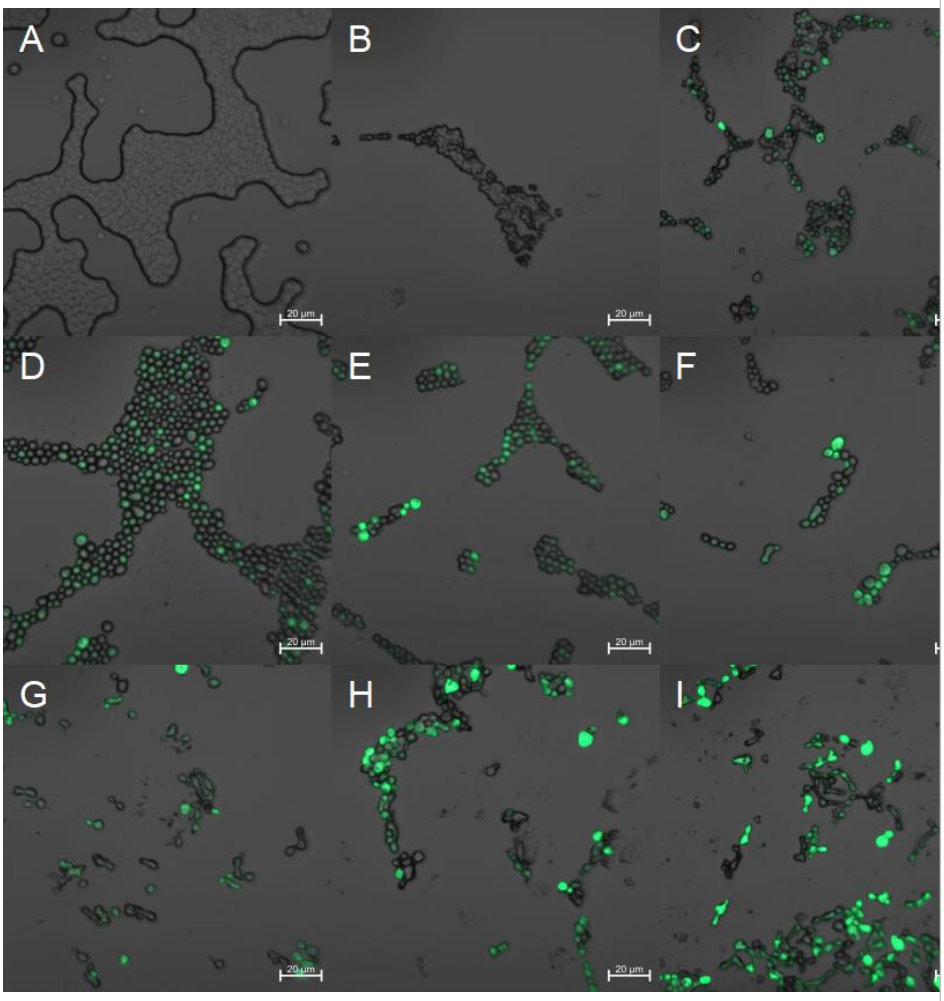
Experiment	Scale-up culture and glycerol preservation of the activated yeast
Note	On 22 July the four yeast cultures activated on 20 July (labelled 3, ③, 4, ④) were taken out and their OD600 was measured at 13:16: 3 = 2.2434, ③ = 2.1118, 4 = 2.0535, ④ = 2.0796. 500 μL of each culture was then inoculated into 45 mL of SD-His liquid medium for expansion (labelled 3, ③, 4, ④) and shaken at 30 C, 200 rpm (started at 13:50). At 18:49 the OD600 was measured again: 3 = 0.7830, ③ = 0.6402, 4 = 0.6584, ④ = 0.6959. Glycerol stock preservation: 0.75 mL of 50% glycerol was mixed with 0.75 mL of the corresponding culture and aliquoted for frozen storage - stocks 1 and 2 from culture 3, stocks 3 and 4 from culture ③, stocks 5 and 6 from culture 4, and stocks 7 and 8 from culture ④; the remaining cultures were returned to 30 C, 200 rpm at 19:15 for continued culture.
Result	The OD600 of the four cultures was 2.0535-2.2434 at 13:16 (before expansion); after 1:90 dilution (500 μL into 45 mL) and about 5 h of shaking culture it read 0.6402-0.7830 at 18:49. A total of 8 glycerol stocks (nos. 1-8) were prepared and stored frozen, and the remaining cultures were kept growing.
Explanation	The yeast expanded and re-preserviced into glycerol stocks in this experiment is the *P_FUS1+yEGFP* fluorescent reporter-strain yeast constructed earlier. Because activating the glycerol stocks had previously required about 2 days each time, a loss of viability in the glycerol stocks was suspected; therefore the four cultures activated on 20 July (labelled 3, 3-circled, 4, 4-circled) were re-preserved as glycerol stocks (nos. 1-8) to rejuvenate the strain and improve subsequent activation efficiency. Before expansion the cells were in a growth plateau (OD600 about 2.0-2.2); after 1:90 dilution into fresh SD-His medium they re-entered exponential growth and read about 0.64-0.78 at 18:49, indicating healthy growth. Each culture was mixed with an equal volume of 50% glycerol (final glycerol about 25%) and stored frozen.
Next plan	The remaining cultures kept growing will be used for subsequent gradient-concentration fluorescence-induction assays and growth monitoring; the newly prepared glycerol stocks (nos. 1-8) will be kept for long-term preservation and to improve the efficiency of future glycerol-stock activation.

No 35 Date: July 23-24 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	Fluorescence detection of the *yEGFP* reporter under α-factor induction
Note	This experiment followed Protocol 16 (Detection of Yeast Fluorescence Signal After Treatment, 96-well gradient α-factor induction) and Protocol 14 (Yeast Growth Curve Measurement, tube-based gradient α-factor induction with OD600 monitoring). The *P_FUS1+yEGFP* reporter yeast was induced with gradient α-factor (24, 12, 6, 3, 1.5, 0.75, 0.375 μM, plus a 0 μM sterile-water/medium negative control); growth was assessed by monitoring OD600 growth curves, and fluorescence was read on a microplate reader in the *yEGFP* green channel (Ex 485 nm / Em 528 nm). Deviation from the protocol: the protocol requires measurement once every hour from induction (growth curve up to about 10 h; fluorescence read continuously for 16 h, 16 time points in total). Owing to the large workload and limited experience, sampling was not strictly hourly this time; only a few time points were measured and summarized to obtain the general trend.
Result	Growth curves: the 0 μM control grew fastest and reached the highest final cell density; as the α-factor concentration increased, both the growth rate and the final biomass decreased in a concentration-dependent manner, with the inhibition most pronounced in the high-concentration groups (6-24 μM). Fluorescence: at a given time point the relative fluorescence increased monotonically with α-factor concentration and gradually saturated, showing a good dose-response relationship; at a given concentration the fluorescence rose with induction time, the increase slowed and plateaued after about 10-16 h, and the fluorescence then began to decline.  **Fig. 1 Growth curves of yeast under gradient α-factor induction (OD600)**

	 **Fig. 2 Fluorescence signal detection under gradient α-factor induction**
Explanation	α-factor, a mating pheromone, arrests S. cerevisiae in G1 and consumes cellular energy, so higher concentrations inhibit growth more strongly. In the reporter the α-factor-responsive promoter *P_FUS1* drives *yEGFP*, so stronger and longer induction yields more *yEGFP*, giving fluorescence that rises and saturates with concentration and time; the later decline reflects entry into stationary/decline phase, deteriorating cell state and fluorescence-protein degradation, indicating that α-factor-induced fluorescence output depends on both time and concentration. As sampling was not performed at every hour, the above are trend-level conclusions based on sparse time points.
Next plan	Next, the plan is to strictly re-measure once every hour as specified by Protocols 16/14 to complete the dose-time response curves, and, if needed, combine confocal fluorescence imaging (Protocol 17) to further verify *yEGFP* expression.

No 36 Date: July 24 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	Confocal validation of the *P_FUS1+yEGFP* reporter response
Note	On 24 July, following the confocal observation method described in No. 20 (40× objective, 488 nm excitation), the yeast strains were induced with a gradient of α-factor and then imaged by laser confocal microscopy to verify, at the single-cell level, the response of the reporter system to α-factor (the mating pheromone). A blank control, a negative control and a panel of positively induced clones were set up: A is the original BY4741 strain not transformed with the fluorescent plasmid, and B is an un-induced (0 μM) negative clone control; both show almost no detectable green fluorescence. Panels C–I are positive clones induced with 0.375, 0.75, 1.5, 3, 6, 12 and 24 μM α-factor, respectively (see Fig. 1).
Result	As the α-factor concentration increased, the intracellular *yEGFP* green fluorescence gradually intensified and became more uniform: A and B were essentially dark, C showed weak green fluorescence, and D–I (0.75–24 μM) displayed progressively stronger and evenly distributed green fluorescence, consistent with the gradient dose–response observed in the plate-reader detection (No. 35).  **Fig. 1 Confocal images of yeast under gradient α-factor induction (40× objective, 488 nm excitation). A: untransformed BY4741; B: un-induced (0**

	μM) negative clone; C–I: clones induced with 0.375, 0.75, 1.5, 3, 6, 12 and 24 μM α-factor.
Explanation	The α-factor, as the mating pheromone, binds the Ste2 receptor and activates the downstream MAPK (pheromone) signaling cascade, which in turn activates the *P_FUS1* promoter and dose-dependently drives expression of the *yEGFP* reporter. The near-zero fluorescence of A (untransformed) and B (un-induced) confirms that the green signal originates from induced *yEGFP* expression rather than cellular autofluorescence, while the progressive increase from C to I shows that α-factor dose-dependently drives reporter expression at the single-cell level.
Next plan	The next step is to perform colony PCR on the yeasts transformed with the previously constructed *P_FUS1+MEL1* reporter system, in order to screen and confirm the colonies that were successfully transformed.

No 37 Date: July 26 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	Colony PCR verification of transformant colonies
Note	On 26 July, following the yeast colony verification PCR method (Protocol 13), 12 single colonies were randomly picked from the SD-His transformation (selection) plates. To retain each picked colony, 2.5 μL of each colony suspension was replica-spotted onto a fresh SD-His selective plate, labeled and incubated inverted to archive the corresponding strain. The remaining cells were resuspended in 10 μL of sterile nuclease-free water and subjected to thermal-cycle lysis (5 cycles of 98 °C for 3 min and 4 °C for 2 min) to release the nucleic-acid template; after brief spinning and vortexing, the supernatant was taken as the PCR template. The verification PCR system (10 μL: 5.0 μL 2× Master Mix, 4.6 μL template supernatant and 0.2 μL each of the 10 μM forward/reverse verification primers) was assembled on ice and run, and the products were verified by agarose gel electrophoresis together with a DNA Marker. The thermal-cycling program (the program shown in the record, transcribed here in text) was: initial denaturation 94 °C for 3 min (1 cycle); cycling 35 times of 94 °C for 30 s, 55 °C for 30 s and 72 °C for 1 min 30 s; final extension 72 °C for 5 min (1 cycle); hold at 16 °C indefinitely. This program is consistent with the standard Taq verification template of Protocol 13 (35 cycles, ~55 °C annealing, ~1 kb/min extension for the ~1.6 kb target), so no parameter deviation was made.
Result	The agarose gel electrophoresis result is shown in Fig. 1: DNA Marker was loaded at the lanes flanking the samples (with 2 kbp and 1 kbp reference bands), and the remaining lanes are the colony PCR products. The majority of the sample lanes show a clear, single specific band of about 1635 bp, indicating successful transformation of the target fragment; the 11th sample lane shows no band at 1635 bp with only a faint non-specific band below it, scored as negative (false positive). The overall result is consistent with the expected transformation-positive pattern.

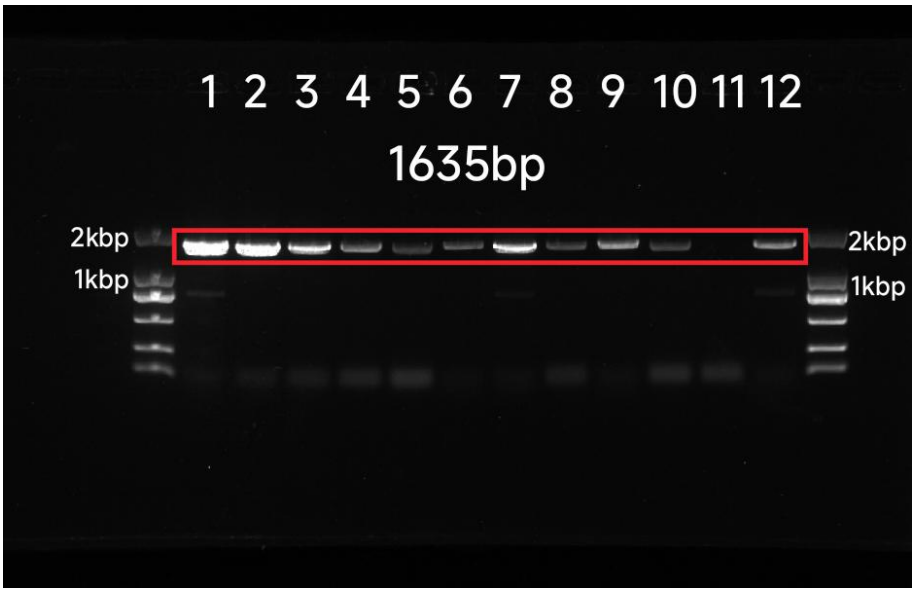
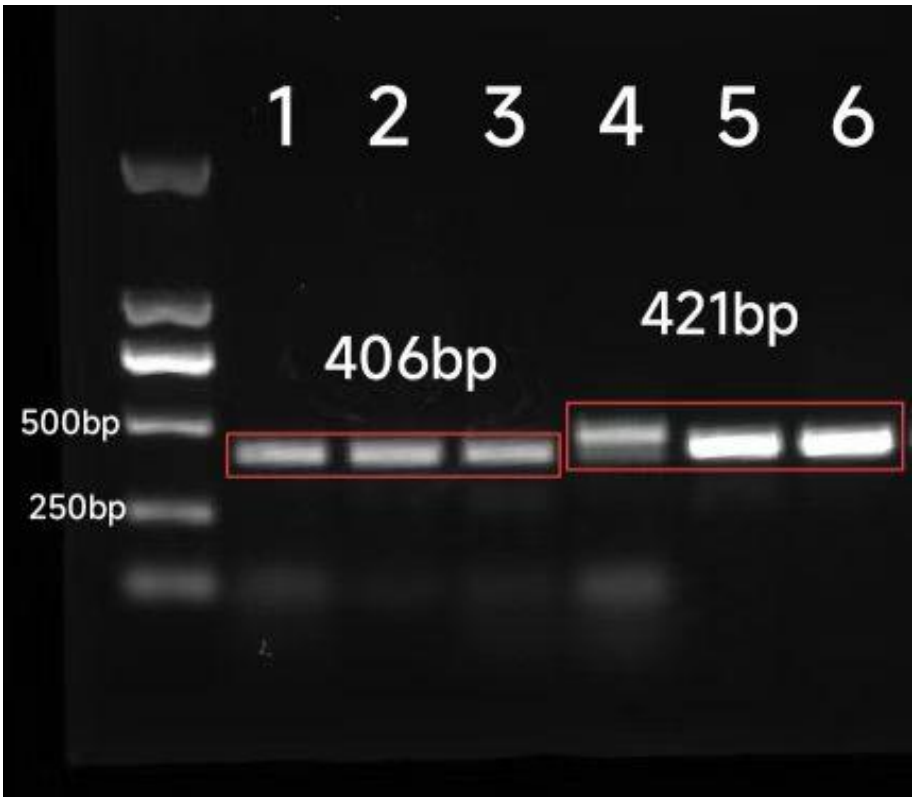


Fig. 1 Colony PCR verification of transformant colonies (specific band of

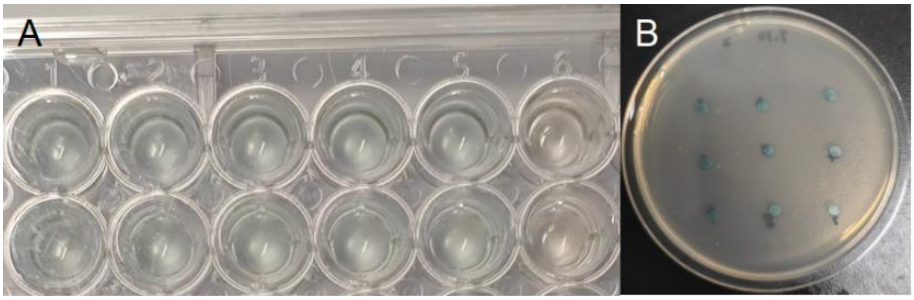
	~1635 bp); M = DNA Marker
Explanation	A single specific band of the expected ~1635 bp size indicates that the target fragment is present in that colony genome, i.e. the colony is a true positive transformant of the constructed *P_FUS1+MEL1* reporter system. The colonies without a band (or with only a non-specific band) are scored negative: possible causes include picking a non-transformant, insufficient cell biomass that failed to release enough template, or incomplete lysis.
Next plan	Perform PCR verification of the two junction sites of the Gibson one-step cloning on the single colonies that were confirmed positive.

No 38 Date: July 28 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	Colony PCR verification of the Gibson cloning junctions
Note	On 28 July, colonies No. 2, 3 and 4 were picked from the 12 single colonies that had been kept previously — the three colonies with the best colony-PCR results in the previous step — and the yeast colony verification PCR method (Protocol 13) was followed to verify the two junction sites of the Gibson one-step cloning. The PCR program was: initial denaturation at 94 °C for 3 min (1 cycle); 94 °C for 30 s, 55 °C for 30 s and 72 °C for 30 s for 35 cycles; final extension at 72 °C for 5 min (1 cycle); hold at 16 °C indefinitely.
Result	The agarose gel electrophoresis result is shown in Fig. 1: DNA Marker was loaded at the flanking lanes (with 500 bp and 250 bp reference bands), and the remaining lanes are the junction PCR products. All three colonies (No. 2, 3 and 4) show two correct and clear target bands of about 406 bp and about 421 bp, corresponding to the two junction sites of the Gibson one-step cloning.  **Fig. 1 Colony PCR verification of the two Gibson junction sites (bands of ~406 bp and ~421 bp)**
Explanation	The presence of both junction amplicons of the correct size (about 406 bp and about 421 bp) in all three colonies indicates that the two junction sites of the Gibson one-step cloning were assembled correctly, confirming that colonies No. 2, 3 and 4 are true recombinant clones.

Next plan	Expand-culture the verified positive colonies and prepare glycerol stocks; the full-length gene may be further confirmed by Sanger sequencing before proceeding to downstream assays.

No 39 Date: July 29-30 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	X-α-gal detection of *MEL1* expression under α-factor induction
Note	Carried out according to Protocol 24. On July 29 the three verified single colonies (no. 2, 3, 4) were inoculated into 5 mL SD-His liquid medium and expanded at 30 °C, 220 rpm from 21:53 on July 29. At 16:16 on July 30 the ODs of the three cultures were 1.8712, 1.7014 and 1.5192, respectively; an appropriate volume of each culture was then diluted to a starting OD of about 0.2 and cultured at 150 rpm, 30 °C from 16:47 to 20:09, giving ODs of 0.3704, 0.4733 and 0.4152. From each tube, 2.5 μL was spotted in triplicate onto SD-His plates containing X-α-gal (Fig. B); the remaining cultures were induced with a gradient concentration of α-factor and X-α-Gal chromogenic substrate was added to the liquid medium (Fig. A).
Result	Fig. A shows the liquid color development: from left to right the wells correspond to the 12, 6, 3 and 1.5 μM α-factor induced groups (columns 1-4), and the degree of blue color development of the liquid medium showed no clear correlation with the induction concentration; column 5 is the negative group without α-factor, whose liquid medium still turned blue, indicating leakage of the system; column 6 is the blank control without cells. Fig. B shows the colonies spotted on SD-His plates containing X-α-gal: after overnight culture the colonies that were not induced with α-factor had already all turned blue, confirming again that the system undergoes chromogenic leakage under the un-induced condition.  **Fig. 1 X-α-gal color development of α-factor gradient-induced *MEL1* expression (A: liquid medium, columns 1-4 = 12, 6, 3, 1.5 μM induction, column 5 = negative control without α-factor, column 6 = blank control without cells; B: colonies spotted on SD-His plates containing X-α-gal)**
Explanation	Both the un-induced negative group in liquid and the spotted colonies turned blue without α-factor, indicating that the *P_FUS1*-MEL1 reporter system exhibits constitutive leaky expression under the basal (un-induced) state, so *MEL1* already hydrolyses X-α-Gal and develops blue color in the absence of induction. The blue intensity of the liquid showed no clear dose-dependence on the α-factor gradient, which may be because the leaky background dominates and masks the induced increment, or is related to the substrate concentration and culture/induction conditions.

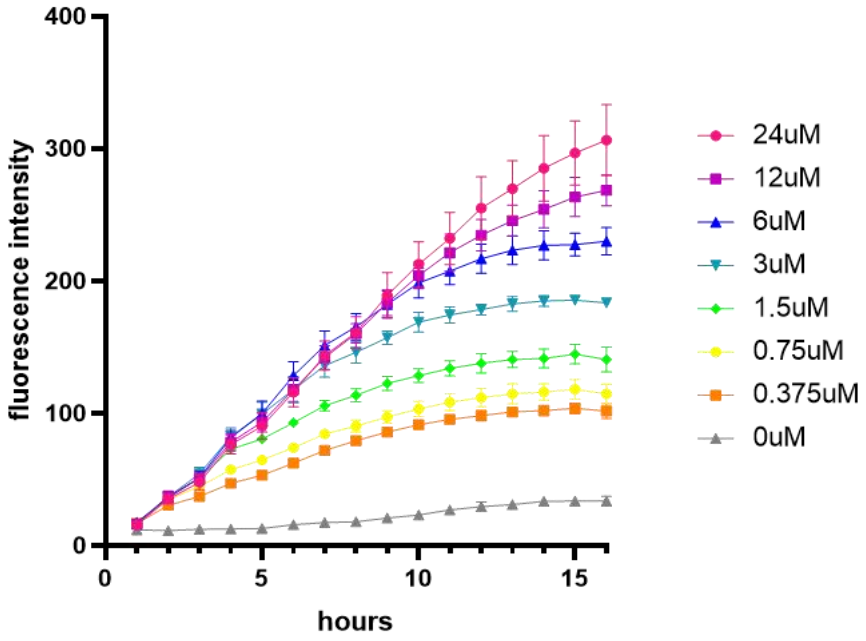
Next plan	Design degenerate primers to mutate the PRE elements of the *FUS1* promoter for mutant-library construction, and screen for non-leaky, inducible strains based on phenotype.

No 40 Date: July 31-Aug 4 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	Sanger sequencing of the *P_FUS1+MEL1* reporter construct
Note	The full-length PCR product obtained on 26 July was sent for Sanger sequencing, and the results were received between 31 July and 4 August (see Figures 1 and 2). The obtained reads (forward and reverse, sample 0001_3332) were aligned against the reference plasmid sequence pESC-HIS-P_FUS1-MEL1 to verify the full-length construct.
Result	Sequence alignment shows that the sequenced reads are essentially identical to the expected reference over the full length, with only a few bases differing - a small number of mismatches and single-base gaps, most of them located at the sequencing ends where the base-call quality drops. The overall sequence is consistent with the design.  **Fig. 1 Sequence alignment between the sequenced full-length product and the reference pESC-HIS-P_FUS1-MEL1 (upstream region)**  **Fig. 2 Sequence alignment between the sequenced full-length product and the reference pESC-HIS-P_FUS1-MEL1 (downstream region)**
Explanation	The few base differences are concentrated at the low-quality sequencing ends and represent sequencing/alignment artifacts or synonymous changes rather than real mutations; they do not change the open reading frame or the key functional regions of *P_FUS1* and *MEL1*. The full-length sequence therefore matches the design, confirming that the *P_FUS1+MEL1* reporter construct has been successfully assembled and can be used for downstream functional validation.

Next plan	The verified correct strain will be expanded and preserved as a glycerol stock; this validated strain will then be used as a template for obtaining gene fragments for *FUS1* promoter optimization.

No 41 Date: Aug 5-6 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	Fluorescence detection under gradient α-factor induction
Note	Carried out according to Protocol 16. On August 5–6, fluorescence intensity was measured under a gradient α-factor induction, with 4 technical replicates set for each group. The induction parameters followed Protocol 16, except for the α-factor final-concentration gradient set as 0, 0.375, 0.75, 1.5, 3, 6, 12 and 24 μM (0 μM as the negative control), and the number of replicates adjusted from 3 to 4 in this experiment (differences noted).
Result	The fluorescence intensity increases with rising α-factor concentration in a dose-dependent manner. The 0 μM negative-control group shows the lowest and nearly flat signal, while all induced groups rise clearly within the detection window, and the 24 μM group shows the strongest response (Fig. 1).  **Fig. 1 Fluorescence intensity of the gradient α-factor induction assay over time (4 technical replicates per group)**
Explanation	The positive correlation between fluorescence and α-factor concentration indicates that the *P_FUS1* reporter responds to the inducer in a dose-dependent manner; the small error bars among replicates suggest good reproducibility of the assay.
Next plan	The *P_FUS1*-MEL1 blue–white reporter system will be further optimized to address its leakage problem.

No 42 Date: Aug 11 Experimenters: Chengxi Luo Recorder: Chengxi Luo

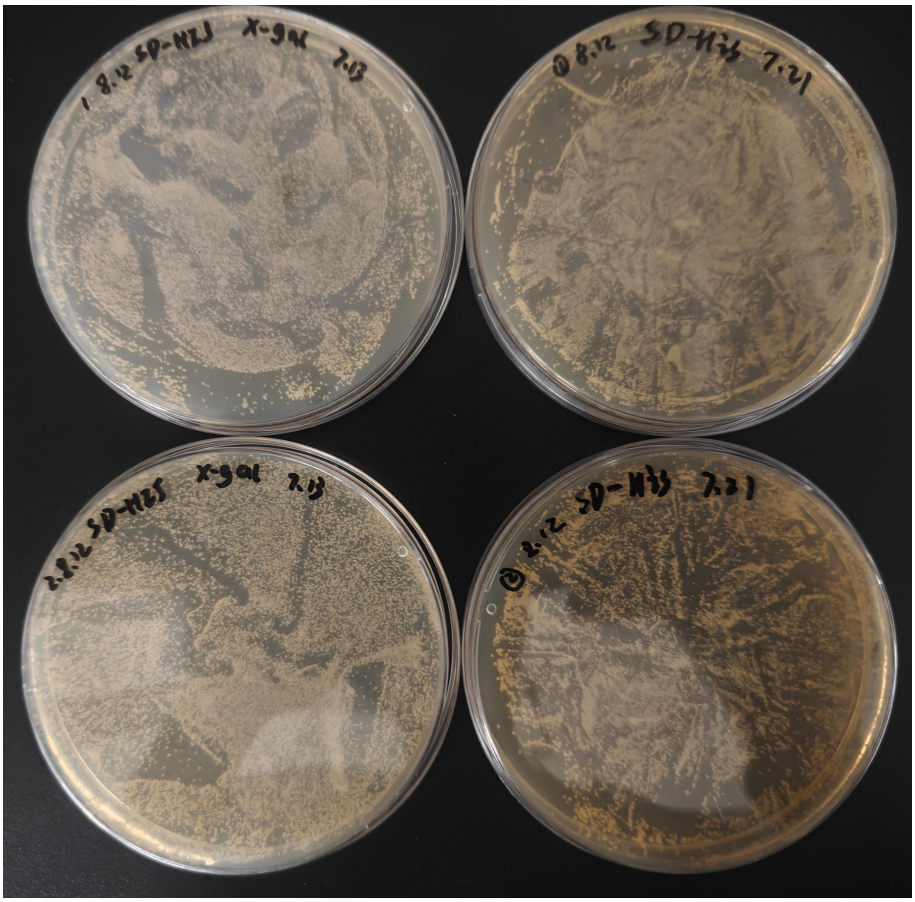
Experiment	PCR of the *PRE1*/*PRE2*-mutant *P_FUS1-MEL1* fragments and vectors
Note	Carried out according to Protocol 13 (target-fragment method). On August 11, the constructed leaky *P_FUS1*-MEL1 reporter yeast was picked and lysed to obtain the genomic DNA template. The designed degenerate primers were used to PCR-amplify the *P_FUS1*-MEL1 fragment carrying mutations in *PRE1* and *PRE2* of the *FUS1* promoter (giving PR1 and PR2), and conventional primers were used to amplify the corresponding linearized vectors (pESC1 and pESC2). Deviations from Protocol 13: degenerate primers (rather than the ordinary colony-PCR primers) were used for the insert to introduce the *PRE1*/*PRE2* mutations, and the same amplification scheme was applied to the vector for linearization. The amplification products were verified by agarose-gel electrophoresis (Fig. 1).
Result	Agarose-gel electrophoresis showed that, apart from one no-band sample for each of the two linearized vectors (pESC1 and pESC2), all other samples gave a single clear and bright band: the mutant fragments PR1 (~1684 bp) and PR2 (~1657 bp) and the remaining linearized-vector lanes pESC1 (~6276 bp) and pESC2 (~6303 bp) were all amplified as expected, confirming successful construction of the *PRE1*/*PRE2*-mutated fragments and their corresponding linearized vectors.



Fig. 1 Agarose-gel electrophoresis of the *PRE1*/*PRE2*-mutated *P_FUS1*-

	MEL1 fragments (PR1, PR2) and their linearized vectors (pESC1, pESC2)
Explanation	Every mutant-fragment lane amplified a single band of the expected size, indicating that the degenerate-primer strategy successfully introduced the *PRE1* and *PRE2* mutations into the *P_FUS1*-MEL1 construct. The two missing bands in pESC1 and pESC2 most likely reflect failed amplification of the vector in those two samples (low template amount, primer-dimer formation, or suboptimal amplification conditions); this does not affect the other lanes, which produced the correctly sized mutant fragments and linearized vectors.
Next plan	Gel-extract and purify each *PRE1*/*PRE2*-mutant fragment and its corresponding linearized vector and transform the constructs into yeast, then screen the transformants for non-leaky, inducible phenotypes by the X-α-gal blue-white assay.

No 43 Date: Aug 12-14 Experimenters: Chengxi Luo Recorder: Chengxi Luo

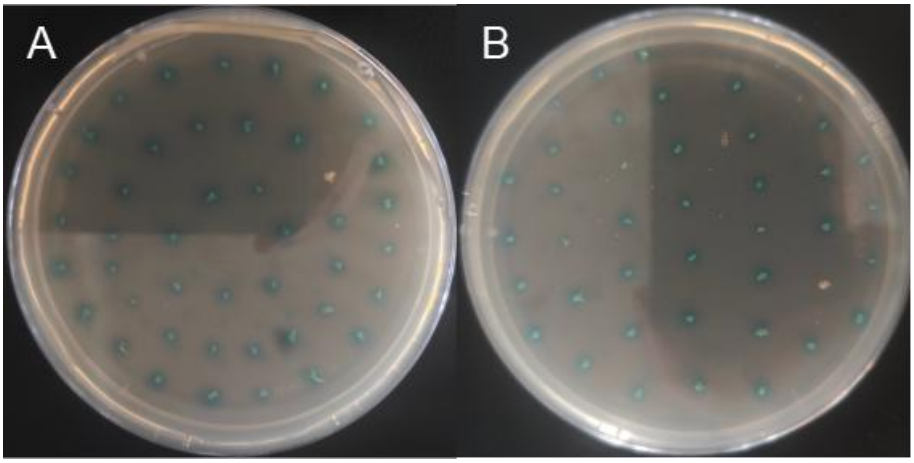
Experiment	Yeast transformation of the *PRE1*/*PRE2*-mutant *P_FUS1-MEL1* fragments
Note	Gel extraction was carried out according to Protocol 10, and yeast competent-cell preparation and transformation according to Protocol 12. On August 12, the previous day's PCR products were gel-purified and mixed at a 1:1 molar ratio, then transformed directly into BY4741 competent cells, relying on the 40-bp homologous arms at both ends of the target fragment for in vivo assembly by yeast homologous recombination. Deviations from the protocols: (i) 40-bp homologous arms were engineered at both ends of the target fragment for yeast in vivo recombination; (ii) no negative control was set, because the operation was well practised and no spare SD-His plates were available.
Result	After plating at 17:55 on August 12 and incubating until 21:00 on August 14, a large number of well-dispersed single colonies grew on the SD-His selection plates (as shown in Fig. 1), indicating that the recombinant plasmid was successfully assembled and transformed.  **Fig. 1 Colony growth on SD-His selection plates after homologous-recombination transformation of the *PRE1*/*PRE2*-mutant fragments and their linearized vectors**
Explanation	The abundant colonies on the SD-His plates indicate that the *PRE1*/*PRE2*-mutant fragments and their linearized vectors assembled efficiently in vivo in BY4741

	and transformed successfully, and that the 40-bp homologous arms were sufficient to support in vivo assembly. Although no negative control was set, the result remains highly reliable given the stability of the preceding experiments and the high transformation efficiency observed here.
Next plan	Pick single colonies for liquid culture and α-factor induction testing, and screen the low-leakage, inducible mutant strains by the X-alpha-gal color-development assay.

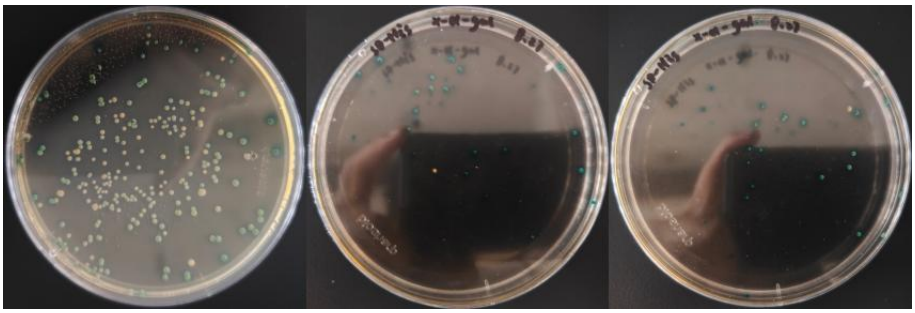
No 44 Date: Aug 15-16 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	96-well liquid blue-white screening of transformants
Note	On August 15, 100 μL of SD-His liquid medium containing the X-α-gal chromogenic substrate and 60 μM (final concentration) α-factor was added to each well of a clear 96-well plate; single colonies were picked from the transformation plates with a sterilized toothpick, transferred into the liquid medium, mixed, and incubated overnight in the dark. On August 16 the same procedure was repeated, but four blank control wells (without cells) were evenly set up in the 96-well plate.
Result	On August 15 all experimental wells turned blue; on August 16 both the experimental wells and the four blank control wells also all turned blue (as shown in Fig. 1).  **Fig. 1 Colour development in the 96-well liquid assay with X-α-gal and 60 μM α-factor (Aug 15-16); the blank control wells also turned blue.**
Explanation	The blank control wells without cells also turned blue, indicating contamination or cross-contamination of the liquid assay system (e.g., aerosol carry-over or toothpick contact), which produced false-positive results; therefore this liquid-based method is not reliable enough.
Next plan	The 96-well liquid colour-development method will be abandoned; single colonies will instead be picked directly onto SD-His solid plates containing the X-α-gal chromogenic substrate for induction screening, in order to reduce contamination risk and improve reliability.

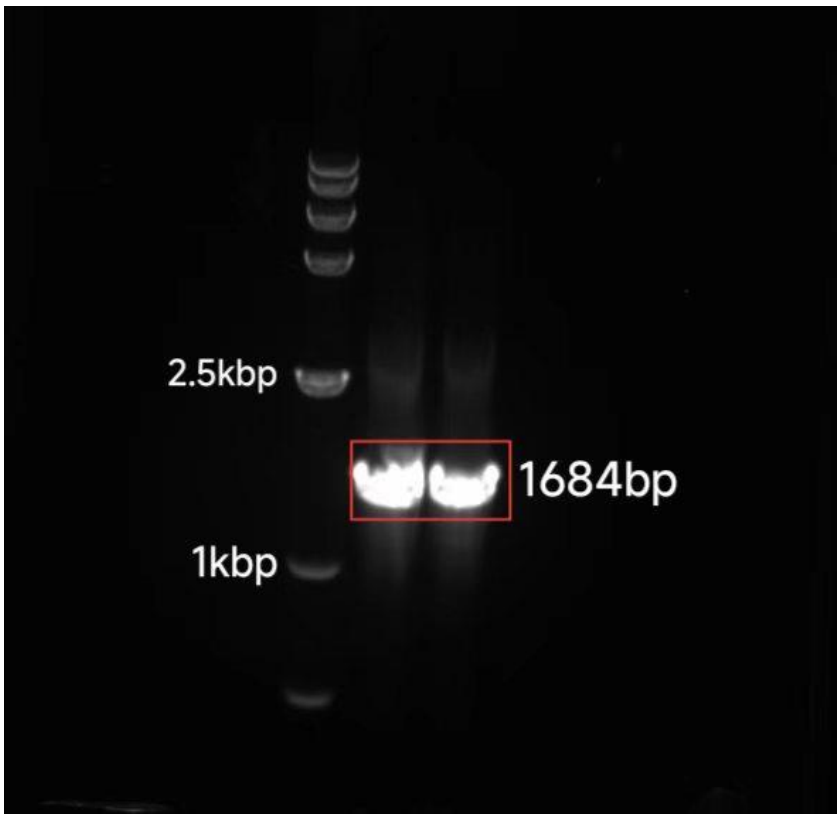
No 45 Date: Aug 17-19 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	Solid-plate blue-white screening and α-factor induction test
Note	On August 17, a sterilized toothpick was used to pick single colonies directly onto SD-His solid plates containing the X-α-gal chromogenic substrate, incubated overnight in the dark until August 18 to obtain blue-white plates (as shown in Fig. 1). On August 18, 2.5 μL of 6 mM α-factor was dropped onto the white single colonies for induction, followed by overnight dark incubation until August 19. Special parameter: induction was performed by directly dropping 2.5 μL of 6 mM α-factor onto the colonies.
Result	On August 18 most colonies on the plate turned blue, indicating leaky expression, while only a very small number of single colonies remained white. After the α-factor induction on August 18 and further dark incubation until August 19, the white colonies still did not turn blue, so no suitable non-leaky, inducible single colony had been obtained (as shown in Fig. 1).  **Fig. 1 Blue-white screening on SD-His solid plates containing X-α-gal after overnight incubation (August 17-18); most colonies are blue with only a few white colonies.**
Explanation	The blue colonies turned blue due to leaky *MEL1* expression, whereas the white colonies that still failed to turn blue after α-factor induction were non-inducible; neither type met the requirement of a non-leaky yet inducible reporter strain. The frequency of candidate colonies was extremely low, making this toothpick-picking method too inefficient.
Next plan	Given the low efficiency of the toothpick-picking method, a dilution spread-plate method will be adopted for screening in subsequent experiments to increase throughput and the chance of isolating non-leaky, inducible single colonies.

No 46 Date: Aug 31-Sep 3 Experimenters: Chengxi Luo Recorder: Chengxi Luo

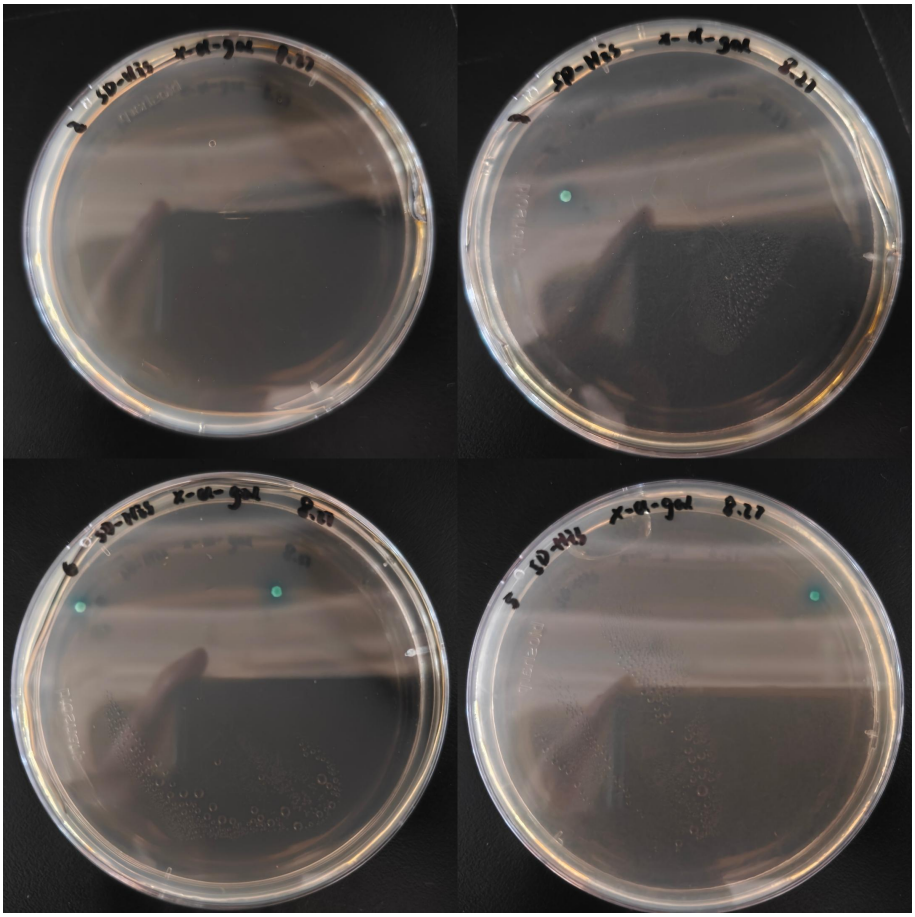
Experiment	Dilution-plating blue-white screening and α-factor induction test
Note	On August 31, the colonies on the No.43 transformation plates were rinsed off with 3 mL of sterile water and mixed thoroughly, then serially diluted by gradient concentration and spread onto SD-His solid plates containing the X-α-gal substrate. After incubation until September 2, colonies from the plate with an appropriate dilution were picked, and the white single colonies were re-streaked onto another plate; those that still did not turn blue were induced by adding 2.5 μL of 6 mM α-factor and incubated in the dark overnight until September 3.
Result	On September 2 the colonies on the dilution spread-plates were well separated and a mixture of blue and white colonies was observed, with the white colonies predominating on the appropriately diluted plates; however, after the white single colonies were induced with α-factor and incubated in the dark overnight until September 3 they still failed to turn blue.  **Fig. 1 Blue-white screening on SD-His solid plates containing the X-α-gal substrate after gradient dilution and spread plating (Aug 31-Sep 2).**
Explanation	White single colonies that still failed to turn blue after α-factor induction lack inducibility, whereas the colonies that turned blue without induction show leaky *MEL1* expression. Consequently, no single colony satisfying both the non-leaky and inducible criteria was obtained even with the dilution spread-plate method, indicating that the leakage problem of the *P_FUS1*-MEL1 reporter system has not yet been resolved.
Next plan	The *P_FUS1* promoter region and the *PRE1*/*PRE2* mutation design will be re-examined to further reduce basal leakage (for example, by adjusting the core promoter elements or the repression sites), and additional mutant libraries will be constructed and screened by the dilution spread-plate method to obtain non-leaky, inducible strains.

No 47 Date: Sep 2 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	*PRE1*/*PRE2* double-mutation PCR of the *FUS1* promoter
Note	The PCR amplification was performed following Protocol 7 (Conventional PCR), and the agarose gel electrophoresis was performed following Protocol 5 (Agarose Gel Electrophoresis); the protocol steps were not repeated here. Special/different parameters: new degenerate primers were designed to introduce the *PRE1*/*PRE2* double mutation at the same time, with the *PRE2*-mutant fragment used as the template, and the template amount was 200 ng. The thermal program was the standard Taq program: 94 °C for 3 min; 28 cycles of 94 °C for 30 s, 55 °C for 30 s and 72 °C for 2 min; final extension at 72 °C for 5 min; hold at 16 °C.
Result	Agarose gel electrophoresis showed that every sample lane contained a single clear and bright target band of the expected size (about 1,684 bp), indicating that the new degenerate primers specifically amplified the *PRE1*/*PRE2* double-mutant *FUS1* promoter fragment.  **Fig. 1 Agarose gel electrophoresis of the *PRE1*/*PRE2* double-mutant *FUS1* promoter fragments amplified by the new degenerate-primer PCR (Sep 2).**
Explanation	The single band at the expected size in every lane shows that the new degenerate primers specifically amplified the target fragment while introducing mutations at both *PRE1* and *PRE2* simultaneously, without producing non-specific products. This double-mutant fragment provides the template for subsequent assembly and screening to address the leakage of the *P_FUS1*-MEL1 reporter system.
Next plan	The double-mutant fragments will be gel-extracted and purified, then assembled

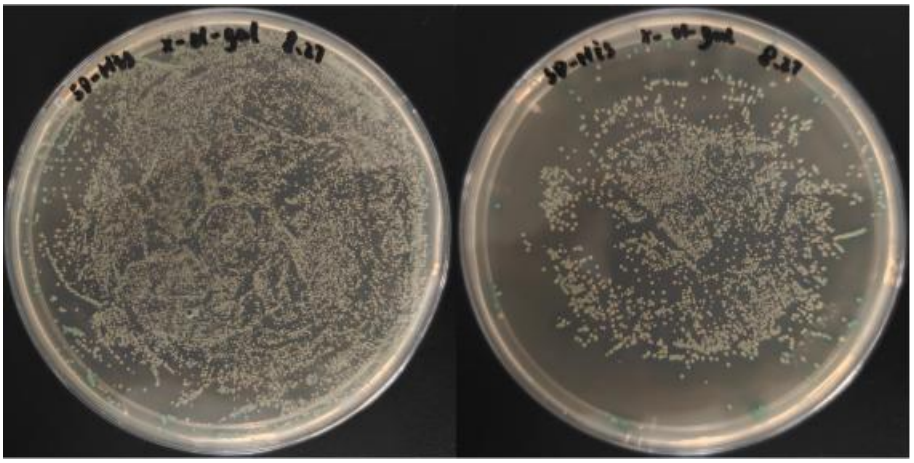
	with the corresponding linearized vectors and transformed into yeast; the transformants will be screened by the X-α-gal blue-white assay combined with α-factor induction to obtain non-leaky, inducible strains.

No 48 Date: Sep 3-9 Experimenters: Chengxi Luo Recorder: Chengxi Luo

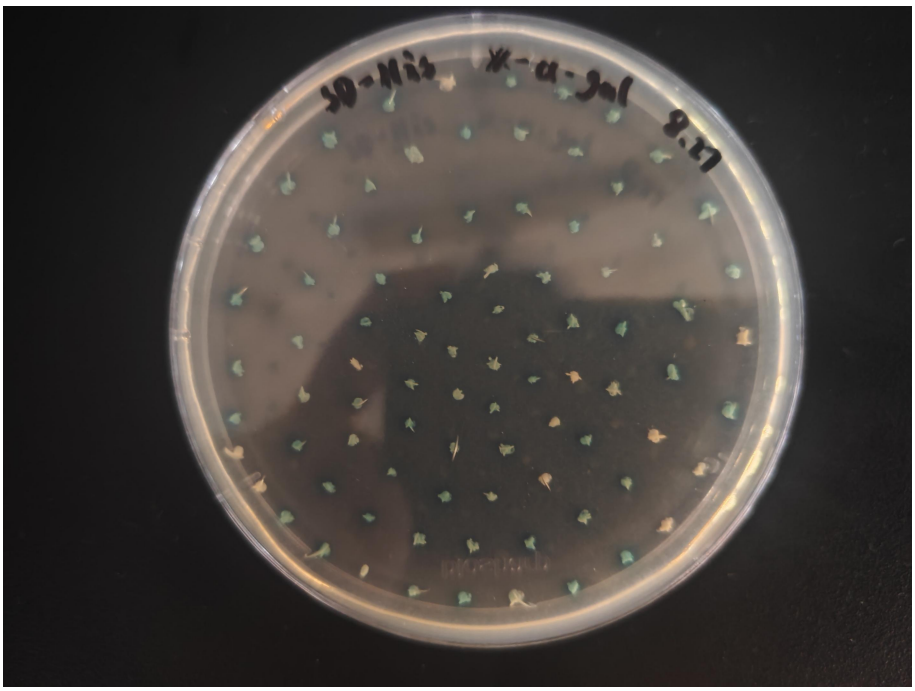
Experiment	Yeast transformation and dilution-plating screening of the *PRE1*/*PRE2*-mutant fragment
Note	Gel extraction was performed following Protocol 10 (Gel Extraction of PCR Products), and yeast competent-cell preparation and transformation were performed following Protocol 12 (Yeast Transformation); the protocol steps were not repeated here. Special/different parameters: the gel-extracted *PRE1*/*PRE2* double-mutant PCR product was mixed with the previously gel-extracted linearized pESC1 vector at a 1:1 ratio and directly transformed into BY4741; after transformation the cells were directly diluted and spread onto SD-His plates. Because the dilution factor was too high, only a few single colonies grew even after incubation until Sep 9.
Result	The transformation mixture was diluted and spread onto SD-His plates containing the X-α-gal substrate. Because the dilution factor was too high, only a few isolated single colonies grew even after incubation until Sep 9; the white colonies that were picked and induced still failed to give any suitable non-leaky, inducible single colony.  **Fig. 1 SD-His screening plates after dilution plating of the transformation (Sep 3-9)**
Explanation	The very low colony number was mainly caused by the excessively high dilution factor, which sharply reduced the number of viable cells plated. Picking and

	inducing these few colonies still yielded no suitable non-leaky, inducible single colony, indicating that the leakage of the *P_FUS1*-MEL1 reporter system was difficult to resolve by screening in the ordinary BY4741 background.
Next plan	The construct will be transformed directly into a G-protein-humanized BY4741 strain, in which the reporter signal is further attenuated, to try to screen out suitable non-leaky, inducible single colonies.

No 49 Date: Sep 11-14 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	Homologous recombination and transformation into BY4741
Note	Competent-cell preparation and yeast transformation following Protocol 12 (Yeast Transformation). On September 11, the previously gel-extracted *PRE1*/*PRE2* double-mutant PCR product was mixed with the linearized pESC1 vector at a 1:1 ratio and transformed directly into the G-protein-humanized BY4741 strain; the mixture was spread onto SD-His plates containing the X-α-gal substrate and incubated inverted at 30 °C until September 14 (Fig. 1).
Result	After incubation until September 14, well-separated single colonies grew on the SD-His plates, and the colonies on both plates were predominantly white (a few faint blue spots appeared on the more densely populated plate), as shown in Fig. 1. White colonies were then picked onto another SD-His solid plate containing the X-α-gal substrate for induction screening of non-leaky, inducible single colonies.  **Fig. 1 SD-His screening plate after transformation into G-protein-humanized BY4741 (Sep 14)**
Explanation	The G-protein-humanized BY4741 background attenuated the reporter signal, and the appearance of numerous white colonies on the SD-His plates indicates weakened leaky expression of *MEL1*, providing candidate colonies for downstream induction verification.
Next plan	Pick the white single colonies onto SD-His solid plates containing the X-α-gal substrate, induce them with α-factor and verify whether they are non-leaky and inducible, so as to identify suitable single colonies.

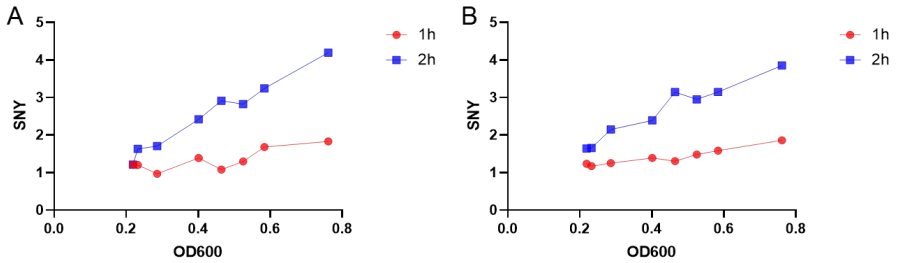
No 50 Date: Sep 14-16 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	Blue-white induction verification of white colonies
Note	On September 14, white colonies from the No.49 transformation plate were picked with a toothpick onto another SD-His solid plate containing the X-α-gal substrate and incubated in the dark overnight until September 15 (Fig. 1). A total of 2.5 μL of 6 mM α-factor was then added to the white colonies that still did not turn blue, and the plate was incubated in the dark overnight until September 16.
Result	After incubation until September 15, the majority of the restreaked colonies turned blue and only a very few remained as white single colonies (Fig. 1). After α-factor induction until September 16, these white colonies still did not turn blue, indicating that suitable non-leaky, inducible single colonies were still not obtained.  **Fig. 1 Blue-white restreak plate after overnight induction in the dark (Sep 15)**
Explanation	The white colonies that failed to turn blue after induction have no inducibility, whereas the blue colonies reflect leaky *MEL1* expression; neither meets the requirement of being non-leaky yet inducible. Colony picking still fails to recover a suitable strain, so a more sensitive readout is needed.
Next plan	A fluorescence reporter system will be used going forward, inducing at different starting OD values to observe the differences in signal change and thereby more sensitively evaluate the induction response and the level of leaky expression.

No 51 Date: Sep 18-20 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	Scale-up culture of the *yEGFP* fluorescent yeast
Note	The expansion culture follows Protocol 25 (*yEGFP* Fluorescent Yeast Induction Assay for Signal Differences at Different Starting ODs); only the parameters that differ are noted here. On September 18, 10 μL of the *yEGFP* fluorescent yeast glycerol stock was inoculated into 5 mL of SD-His liquid medium and activated by expansion culture until September 20, with two tubes prepared in total, to provide cells for the subsequent fluorescence assay examining signal differences at different starting ODs.
Result	The two tubes of *yEGFP* fluorescent yeast grew well in the SD-His liquid medium after expansion, remained in normal growth state and reached the density required for the subsequent experiment.
Explanation	Expanding the culture from the glycerol stock to a sufficient density yields actively growing cells with a consistent physiological status, which is the prerequisite for accurately setting different starting ODs and reliably comparing the differences in induction signal within the fluorescence reporter system.
Next plan	The expanded cultures will be diluted to a starting OD600 of 0.1–0.2 and grown to about 0.4, then dispensed into a black 96-well plate and induced with α-factor at different final concentrations, recording the signal-to-noise ratio against the starting OD and induction time following Protocol 25.

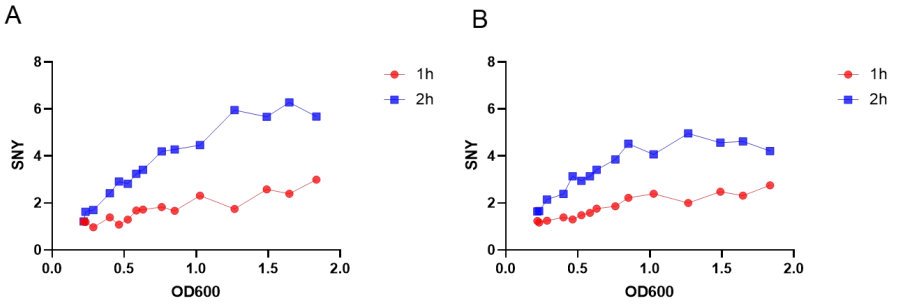
No 52 Date: Sep 20-21 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	Induction-signal detection of the *yEGFP* yeast at different starting ODs
Note	The experiment followed Protocol 25 (*yEGFP* Fluorescent Yeast Induction Assay for Signal Differences at Different Starting ODs) without restating the procedure. Special parameters differing from the protocol are noted: the induction starting OD600 was 0.2190, and only the summary figures of the 12 μM (Fig. A) and 1.2 μM (Fig. B) α-factor induction groups are shown.
Result	Both Fig. A (12 μM) and Fig. B (1.2 μM) show that the signal-to-noise ratio (S/N) increases with rising OD600, and the 2 h induction group generally gives higher S/N values than the 1 h group; at the same OD, the 12 μM induction group yields a slightly higher S/N response than the 1.2 μM group.  **Fig. 1 Summary of signal-to-noise ratio (S/N) for the 12 μM (A) and 1.2 μM (B) α-factor induction groups at different starting ODs**
Explanation	The data indicate that the *yEGFP* fluorescent signal is positively correlated with cell density (OD) and induction time, and a higher concentration of α-factor more effectively triggers the fluorescent response, confirming that this fluorescent system exhibits observable signal-gradient differences at different starting ODs.
Next plan	Based on the current OD–S/N relationship curve, an optimal starting-OD window (e.g. 0.4–0.6) will be selected for subsequent repeat validation.

No 53 Date: Sep 25-27 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	Scale-up culture of the *yEGFP* fluorescent yeast
Note	The expansion culture follows Protocol 25 (*yEGFP* Fluorescent Yeast Induction Assay for Signal Differences at Different Starting ODs); only the parameters that differ are noted here. On September 25, 10 μL of the *yEGFP* fluorescent yeast glycerol stock was inoculated into 5 mL of SD-His liquid medium and activated by expansion culture until September 27, with two tubes prepared in total, to provide cells for the subsequent fluorescence assay examining signal differences at different starting ODs.
Result	The two tubes of *yEGFP* fluorescent yeast grew well in the SD-His liquid medium after expansion, remained in normal growth state and reached the density required for the subsequent experiment.
Explanation	Expanding the culture from the glycerol stock to a sufficient density yields actively growing cells with a consistent physiological status, which is the prerequisite for accurately setting different starting ODs and reliably comparing the differences in induction signal within the fluorescence reporter system.
Next plan	The expanded cultures will be diluted to a starting OD600 of 0.1–0.2 and grown to about 0.6, then dispensed into a black 96-well plate and induced with α-factor at different final concentrations, recording the signal-to-noise ratio against the starting OD and induction time following Protocol 25.

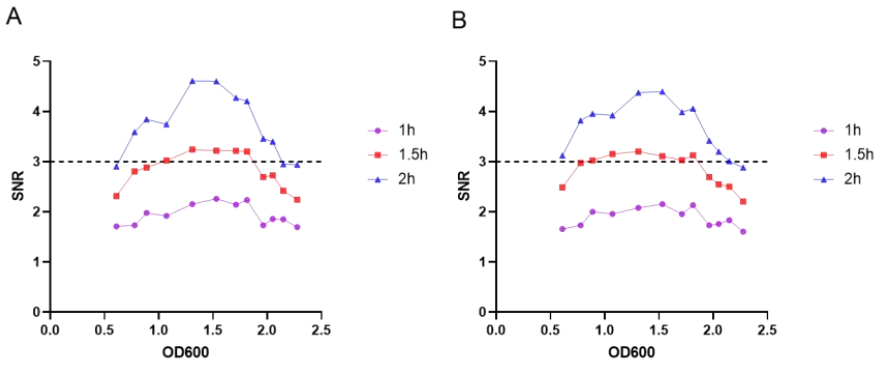
No 54 Date: Sep 27 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	Induction-signal detection of the *yEGFP* yeast at different starting ODs
Note	The experiment followed Protocol 25 (*yEGFP* Fluorescent Yeast Induction Assay for Signal Differences at Different Starting ODs) without restating the procedure. Special parameters differing from the protocol are noted: the induction starting OD600 was 0.2190; the present data were pooled with those of No.52 for a combined summary, and only the summary figures of the 12 μM (Fig. A) and 1.2 μM (Fig. B) α-factor induction groups are shown.
Result	Both Fig. A (12 μM) and Fig. B (1.2 μM) show that the signal-to-noise ratio (S/N) increases with rising OD600, and the 2 h induction group generally gives higher S/N values than the 1 h group; at the same OD, the 12 μM induction group yields a slightly higher S/N response than the 1.2 μM group.  **Fig. 1 Summary of signal-to-noise ratio (S/N) for the 12 μM (A) and 1.2 μM (B) α-factor induction groups at different starting ODs**
Explanation	The data indicate that the *yEGFP* fluorescent signal is positively correlated with cell density (OD) and induction time, and a higher concentration of α-factor more effectively triggers the fluorescent response, confirming that this fluorescent system exhibits observable signal-gradient differences at different starting ODs.
Next plan	Subsequently, the signal-amplified fluorescent yeast with *Far1* and *Sst2* knocked out will be used for the induction assay at different starting ODs.

No 55 Date: Sep 30-Oct 1 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	Scale-up culture of the *Far1*/*Sst2*-knockout fluorescent yeast colony
Note	The expansion culture followed the same procedure as No.51 and Protocol 25 (*yEGFP* Fluorescent Yeast Induction Assay for Signal Differences at Different Starting ODs), so the procedure is not restated; only the parameters that differ are noted here. On September 30, a single colony of the signal-amplified fluorescent yeast with *Far1* and *Sst2* knocked out was picked and inoculated into 5 mL of SD-His liquid medium for expansion culture until October 1, with two tubes prepared in total, to provide cells for the subsequent fluorescence starting-OD induction assay.
Result	The single colony of the knockout strain grew well in the SD-His liquid medium after expansion, remained in normal growth state and reached the density required for the subsequent experiment (no photograph was taken in this entry).
Explanation	Expanding from a single colony of the signal-amplified strain yields a monoclonal, actively growing culture with a consistent physiological status, which is the prerequisite for accurately setting different starting ODs and reliably evaluating the signal amplification effect of the *Far1*/*Sst2*-knockout strain against the background of Protocol 25.
Next plan	The expanded culture will be diluted to a starting OD600 of 0.1–0.2 and grown to about 0.6, then dispensed into a black 96-well plate and induced with α-factor at different final concentrations, recording the signal-to-noise ratio against the starting induction OD and induction time following Protocol 25.

No 56 Date: Oct 1-2 Experimenters: Chengxi Luo Recorder: Chengxi Luo

Experiment	Induction-signal assay of the *Far1*/*Sst2*-knockout yeast at OD600 0.6093
Note	The assay followed Protocol 25 (*yEGFP* Fluorescent Yeast Induction Assay for Signal Differences at Different Starting ODs) and used the *Far1*/*Sst2*-knockout signal-amplified strain expanded in No.55, so the procedure is not repeated here. The special differing parameter is noted: the starting induction OD600 was 0.6093, and only the 12 μM (Fig. A) and 1.2 μM (B) α-factor induction groups are shown.
Result	Both Fig. A (12 μM) and Fig. B (1.2 μM) plot the signal-to-noise ratio (SNR) against OD600 for the 1 h, 1.5 h and 2 h inductions, with a dashed reference line at SNR = 3. The results show that, within a starting OD600 of about 0.6–2.1, the SNR is generally able to reach above 3, i.e. a clear signal can be reliably produced within 2 h. The 2 h group gives the highest SNR (peaking around OD600 1.4–1.6), whereas the 1 h group stays below the SNR = 3 threshold throughout.  **Fig. 1 Summary of signal-to-noise ratio (SNR) for the 12 μM (A) and 1.2 μM (B) α-factor induction groups at different starting ODs (starting induction OD600 = 0.6093).**
Explanation	The *Far1*/*Sst2*-knockout signal amplification widens the usable OD window and raises the SNR, so that over a starting OD600 of roughly 0.6–2.1 a detectable response (SNR ≥ 3) is reached within 2 h. Longer induction further strengthens the signal, indicating that the amplified system performs stably across a wide range of starting ODs.
Next plan	The yeast chassis will be engineered to try to eliminate the signal leakage of the *P_FUS1* + *MEL1* system.